

Computer In Everyday Life

A course in COMPUTER SCIENCE
and INFORMATION TECHNOLOGY

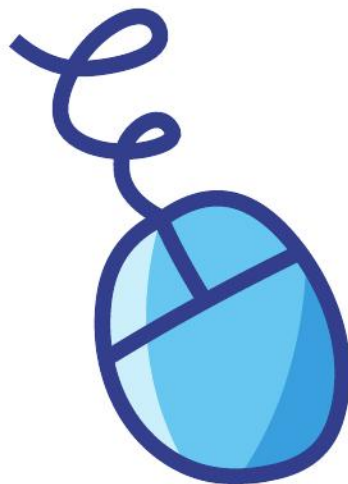


Class

Authors:

Archana Bhogle
M.Sc. (Comp. Sc.), MCA

Kanika Dubey
B.Sc., MCA



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Written by:

Archana Bhogle
Kanika Dubey

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Preface . . .

Today, we are living in the age of Digital Technology. The rapid growth in the field of information and communication technology has completely transformed human life. The impact of technology in our modern-day life is such that it has become more of a necessity for us rather a facility. Development of smaller but powerful electronic gadgets like tablets, smartphones, kindles etc. have made it possible to access any type of service from anywhere using the online mode.

Hence, in the modern era study of information and communication technology is a necessity for each one of us. This study should start early in life so that the new generation can be empowered with this technological advancement to form the initial stage of their learning process.

Computer In Everyday Life a series of eight books is our humble effort to provide the latest, comprehensive learning and understanding of Computer Science. The series conforms to the National Curriculum Framework (NCF) 2005. Interspersed with illustrations, tips and interesting facts, the concepts have been presented in a lucid and student-friendly manner. We have tried to include the latest versions of all software and introduced some new interesting software that is open source and can be procured easily.

The focus of books 1 to 5 of **Computer In Everyday Life** is on learning the basics of computers and to develop skills to operate computers in a competently and efficient manner. Books of level 1 and 2 use lots of illustrations and software such as Tux Paint and Paint to infuse students interest in the subject. Books 3 to 5 teach the basics of Windows 7, Microsoft Office 2010 package and introduce the students to the world of programming through LOGO. The progression of the series duly takes care of comprehension level at each stage and continuity has been maintained while moving from one level to another.

Books 6 to 8 take the students to a different realm where they are encouraged to explore and experiment with software such as Adobe Flash CS3 and Adobe Photoshop CS3. The foundation of programming is further strengthened through chapters on QBasic, HTML and Visual Basic.

We sincerely hope that this series of computer books will help the young minds to equip them with the required IT skills at different stages of their learning process. Your valuable feedback and suggestions will be highly appreciated for improving the areas which have gone unnoticed in these books.

—Authors



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Computer's Memory

This chapter includes



- » Measuring the Memory of the Computer
- » Silicon Chips
- » Computer Memory

Human beings use their memory to store information. That's how we remember things when we need to. Similarly, the computer also has memory, which it uses to store data and instructions. It can bring back information from its memory as and when required.

Measuring the Memory of the Computer

The memory of the computer is measured in units of **bits** and **bytes**.

The full form of bit is **binary digit**. It is the smallest unit of memory. It can have a single numeric value of 0 or 1.



Fact Bytes



We need 1 byte of memory to store one letter and 2 bytes to store a number.

8 bits placed together form 1 byte.

8 Bits	=	1 Byte
1024 Bytes	=	1 Kilobyte (KB)
1024 Kilobytes	=	1 Megabyte (MB)
1024 Megabytes	=	1 Gigabyte (GB)
1024 Gigabytes	=	1 Terabyte (TB)
1024 Terabytes	=	1 Petabyte (PB)



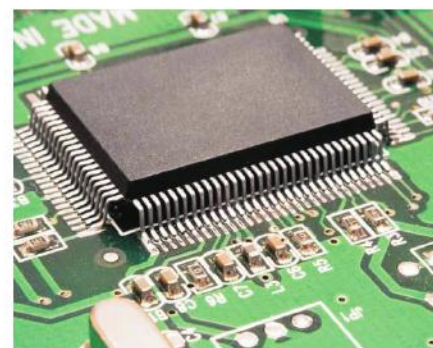
Example :

Convert the following into their byte equivalents.

- (i) 240 KB = 240×1024 bytes
- (ii) 4.2 TB = 4.2×1024 GB
= $4.2 \times 1024 \times 1024$ MB
= $4.2 \times 1024 \times 1024 \times 1024$ KB
= $4.2 \times 1024 \times 1024 \times 1024 \times 1024$ bytes
- (iii) 1.44 MB = 1.44×1024 KB
= $1.44 \times 1024 \times 1024$ bytes
- (iv) 20 GB = 20×1024 MB
= $20 \times 1024 \times 1024$ KB
= $20 \times 1024 \times 1024 \times 1024$ bytes

Silicon Chips

The electronic circuits carrying bits are stored on chips. These chips are tiny pieces of a material called **silicon**. Each chip is covered with lakhs of circuits. All the chips in a computer are mounted on boards called **Printed Circuit Boards (PCBs)** or **daughterboards**. These daughterboards are attached to a motherboard.



Fact Bytes



The silicon chip was invented by two American electrical engineers, Robert Noyce and Jack Kilby. This invention changed the way computers worked and led to the development of modern computers.



Computer Memory

The computer stores all the work done on it. The place where information is stored is called the **memory** of the computer. Computer memory is of two kinds, **primary memory** and **secondary memory**.

1. Primary memory

The primary memory of the computer tells us the amount of data that the computer can store. There are two kinds of primary memory, **RAM** and **ROM**. They come in the form of chips.



Random Access Memory (RAM)

Random Access Memory or **RAM** is also called **read/write memory**. This memory is very small and limited. So we cannot store large information in this memory. This memory stores all the information in the form of electric signals. So whenever power fails or you switch off the computer, all the information that has been stored is lost. This means it is volatile. Therefore, there was a need of some other memory medium where a large amount of data and information can be stored permanently.



RAM

Fact Bytes



It is called RAM because when the computer process data, it can randomly access or pick up any piece of data stored on the RAM.

RAM is of two types:

(i) SRAM (Static RAM): It uses flip-flops as storage elements. In SRAM, once data is written into memory location, the data stays unchanged unless some new data is entered into that location.

(ii) DRAM (Dynamic RAM): It uses capacitors as storage elements. The dynamic RAM content may change with time due to leakage of charge. So it is required to refresh the storage elements periodically. It consumes less power as compared to SRAM.

Read Only Memory (ROM)

It refers to **Read Only Memory**. It usually stores various programs and instructions needed to run those programs. Those programs tell the computer how to work. For example, when the computer is booting up, instructions from the ROM tell the CPU how to start. It is called 'Read Only' because programs stored on the ROM can only be read. One cannot write on the ROM. Unlike the RAM, the ROM holds data permanently. The information on the ROM does not get wiped out when the computer is turned off.



ROM



There are three kinds of ROM :

- (i) **PROM (Programmable Read Only Memory)**: It uses fusible links that can be burned by using special PROM burning circuit. It can be programmed by the user using a PROM programmer.
- (ii) **EPROM (Erasable Programmable Read Only Memory)**: The information stored in an EPROM can be erased by exposing the memory to ultraviolet light.
- (iii) **EEPROM (Electrically Erasable Programmable Read Only Memory)**: It is also same as EPROM but the erasing is done by electrical signals.

2. Secondary Memory

Sometimes, there is so much information that it cannot be saved on the primary storage devices. In that case, data can be saved outside the computer on a secondary storage device. This is called secondary memory. Also, data may be saved for later use. In such a situation, we need to store information on a secondary memory device. A few examples of secondary memory or storage devices are pen drives, hard disks, CDs, DVDs, and floppy disks. This is also called auxiliary memory or backup memory.



Pen drive



Floppy disk



CD



Hard disk

Secondary memory devices

The amount of data that can be stored on any of these devices is called its capacity. It is measured in megabytes or MB, (1 MB = 1024 KB) and gigabytes or GB (1 GB = 1024 MB).

CD

It stands for **Compact Disk**. It is a round shiny disk that can store and read data and programs. There is a CD drive in a CPU to run a CD. A CD can hold a lot of data (650 MB or more).

DVD

It stands for **Digital Versatile Disk**. It is a flat round disk, the same size as a CD. It has a 4.7 inch diameter and is 0.05 inches thick. It can store a lot more data than a CD. The capacity of different types of DVDs varies from 2 GB to 17 GB. They are used mainly for storing films and multimedia files. DVDs require a separate drive in the CPU. A CD can be run in DVD drives, but a CD drive cannot run a DVD. CDs and DVDs are also known as optical disks.



Hard Disk

A **hard disk** is also called a **fixed disk**. It can hold more information than a CD or DVD and gives information much faster. The hard disk is placed inside the CPU. You cannot remove or even see it. Whatever you save in different programs gets saved on the hard disk—unless another disk is specified.

The surface of a hard disk is coated with a magnetic metal. A small recording head is placed very close to its surface to read and write. Data on the hard disk remains there until you delete it.



Pen Drive

A pen drive is a small, portable storage device, which is very different from a hard disk or a CD. We can connect pen drives to our computer through USB ports.

Storage capacities of pen drives vary from 16 MB to almost 1 TB. Unlike CDs and DVDs, pen drives can be used to write and rewrite your work many times, making them better than CDs for secondary storage. A pen drive is also known as a USB drive or a flash drive. A pen drive or a USB drive is inserted into a special slot on the CPU. This slot is called a USB port.

Nowadays many devices such as tablets, laptops, palmtops and music players have USB ports built on them.



Memory Card

A **memory card** or **flash card** is an electronic flash memory data storage device used in many electronic devices, including digital cameras, mobile phones, laptop computers, MP3 players, and video game consoles. They are small, re-recordable and are able to retain data without power. They are commonly used for storing video and data.



Let's Summarise



- ▶ The place where information is stored is called memory.
- ▶ Memory is of two kinds–Primary and Secondary.
- ▶ Primary memory is divided into two types–RAM and ROM.
- ▶ RAM is also called read/write memory.
- ▶ ROM keeps data permanently.
- ▶ Some common secondary memory storage devices are–hard disk, CD, DVD.
- ▶ A hard disk is also called a fixed disk and is made of a metal plate fixed on a spindle.
- ▶ Compact disks are known as CDs.
- ▶ Pen drive is attached outside the computer at a place called USB port.



Practice Time

A. Tick (✓) the correct option.

1. Shortest form of binary digit is _____.
 (a) Byte ☐ (b) Bit ☐ (c) KB ☐
2. _____ is an example of secondary memory.
 (a) RAM ☐ (b) ROM ☐ (c) Hard Disk ☐
3. 1 KB = _____ Bytes.
 (a) 1024 ☐ (b) 124 ☐ (c) 8 ☐
4. The form of bit is _____.
 (a) Bytes ☐ (b) Binary Digit ☐ (c) None ☐
5. 1 GB = _____ MB.
 (a) 124 ☐ (b) 1024 ☐ (c) 8 ☐

B. Write T for true and F for false statements.

1. The place where information is stored is called memory.
2. Memory are of three types.
3. RAM is a primary memory.
4. Compact Disk is the full form of CD.
5. Flash drives are also known as Pen drives.



C. Fill in the blanks.

1. All data in computer get stored and processed as _____.
2. _____ is a permanent memory.
3. _____ is the full form of DVD.
4. _____ is commonly used for storing video and data.
5. _____ is a fixed disk.

D. Answer the following questions.

1. Differentiate between Bit and Bytes.

2. What are silicon chips?

3. What is the use of primary memory?

4. Why is secondary memory important for computers?

5. State the difference between Pen drive and Memory Card.

Fun Time

- Identify the following devices and write their names.













Lab Time



- Visit your computer lab and prepare a list of secondary devices present in your lab.

Brain Booster



- A computer stores information in the form of 0 and 1 . Does it mean a computer file is actually a combination of 0 and 1? Yes or No.

Project Time



- Visit a computer hardware store and collect information on the latest secondary storage devices. Make a comparative chart.

Teacher's Note : Take the students to the computer lab and show them all the secondary memory devices.



Computer Software and its Types

2.



This chapter includes



- ▶ Hardware
- ▶ Types of Software

- ▶ Software

A computer system consists of two main components, **hardware** and **software**.

Hardware

The physical components of a computer system that we can see and touch are called **hardware**.

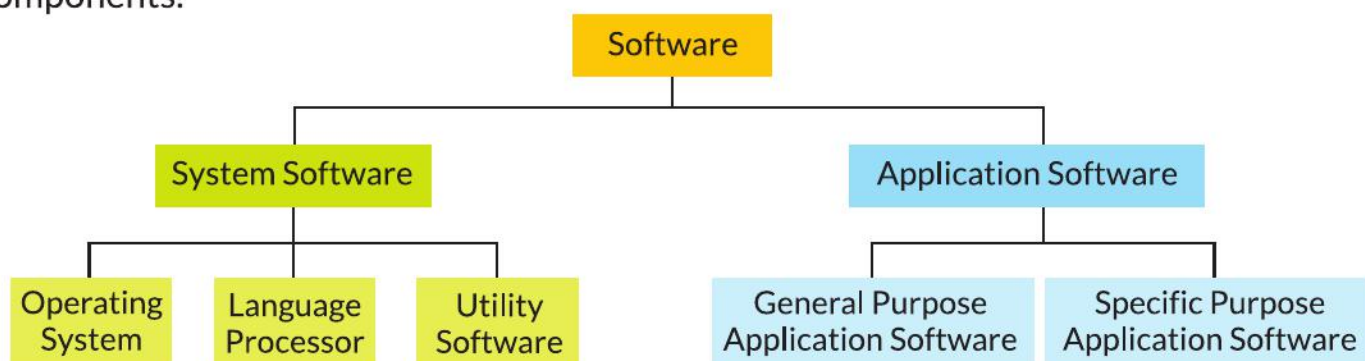
Input and output devices, storage devices, and the central processing unit are examples of hardware.

Software

The term **software** refers to the computer programs that are needed for running or operating a computer system. A program is a sequence of step-by-step instructions to perform a particular task.

Software is an essential component of a functional computer system. Computer hardware is useless without software. Software commands the hardware as to what to do, how to do, and when to do.

You cannot touch software as you can do with hardware. The games you play on a computer, the music and video files that you listen to and see on a computer, the documents on which you work and most importantly, the operating system environment in which you work on your computer system make up the software components.



Types of Software

Software can be classified into the following broad categories:

System Software

The **system software** is required to control the overall operations of a computer system. It can be classified into three categories–Operating System, Language Processors and Utility Software.

Operating System

An **Operating System (OS)** is the master control program that runs the computer. It controls the flow of signals from the central processing unit to the various parts of a computer. When the computer is switched on, the operating system is the first program that gets loaded in its memory.

Some of the basic functions that an operating system performs are:

- ▶ It provides an environment in which computer hardware components are used in an efficient manner.
- ▶ It acts as a resource manager. A computer system has many resources such as CPU time, memory space and I/O devices. The operating system allocates these resources to specific programs and applications as and when required.
- ▶ It lets you create and manage files and folders for storing information in a computer efficiently.

Some examples of OS are–Microsoft Windows, GNU Linux and Unix. Of these, Microsoft Windows is the most popular operating system.



Logos of some operating systems

Fact Bytes



Software piracy is the unauthorised copying or distribution of copyrighted software. We should always use licensed software.

Language Processor

Software are created by writing programs in computer languages. There are many computer languages such as C, C++ and Java. The programs written in these languages cannot be understood by a computer directly. A computer is an electronic machine and can only understand the machine language which comprises 'zeroes' and 'ones'.

A **language processor** is a very important software as it translates or converts a program written in a computer language into the machine language so that it can be understood by a computer.



Utility Software

Utility Software is a program designed to perform maintenance or housekeeping work on a computer system in order to ensure its smooth functioning. Examples of this type of software are–Disk Defragmenter, Antivirus software and Backup Utility.

- ▶ **Disk Defragmenter.** There may be some files whose contents are scattered across different locations on a hard disk. This may slow down the speed at which data can be accessed from such files. Disk defragmenter speeds up the access of data from a hard disk by rearranging the files and the free space. Files are stored in nearby locations and the free space is combined together to form contiguous blocks with the help of disk defragmenter.
- ▶ **Antivirus Software.** A virus is a computer program that may disrupt the normal functioning of your computer. It can slow down the working of a computer or may sometimes even lead to system crash. In order to remove such dangerous programs, antivirus programs need to be used. An antivirus program scans your disk for viruses and removes them. Few examples of antivirus software are–Norton, AVG, Avira and McAfee.



Logos of some antivirus software

- ▶ **Backup Utility.** This software program lets you maintain a copy of the data on the hard disk so that in case of any accidental loss or system crash, valuable data is not permanently lost.

Application Software

System and utility class of software relate more to the computers themselves rather than the users.

An **application software** is a set of programs designed to help users perform specific tasks. They may be installed on a computer depending upon the user requirements. Different application software are available to serve different purposes. The application software can further be divided into two categories–General Purpose Application Software and Specific Purpose Application Software.

General Purpose Application Software

General purpose application software are developed keeping in mind the general needs and requirements of a large number of people. Some examples of such software are:

- ▶ **Word-processing Software.** This type of software is used for creating text-based documents such as letters, stories and reports. All word-processing software



provide options for creating, editing, formatting and printing text and graphics in a document. Examples of word-processing software are–Microsoft Word, Word Perfect, OpenOffice.org Writer, etc.



Logos of some word-processing software

- **Spreadsheet Software.** This software accepts data values in row and column format to allow users to manipulate, calculate and analyse data in the desired manner. Examples of spreadsheet software are–Excel, Lotus 1-2-3, OpenOffice.org Calc, etc.



Logos of some spreadsheet software

- **DBMS (Database Management System).** DBMS is a software program that lets you store a large amount of data in an organised manner. A DBMS can store details about the customers, employees, products, students, etc. Examples of DBMS software are–Oracle, Microsoft Access, FoxPro, OpenOffice.org Base, etc.



Logos of some DBMS software

- **DTP (Desktop Publishing Software).** This software is used for creating designs and layouts of books, journals, brochures, newspapers, etc. Examples of Desktop Publishing Software are–QuarkXPress and Adobe InDesign.



Logos of some desktop publishing software



- **Graphics Software.** This type of software allows you to create and edit images. Examples of graphics software include Adobe Photoshop, CorelDraw, etc.
- **Multimedia Software.** This software lets you combine images, text, sound, animations and videos to create interactive applications. Examples of multimedia software are–Adobe Director, Adobe Flash, etc.
- **Presentation Software.** This software application is used to create and design slide shows and multimedia presentations. Examples of this category of software are Microsoft PowerPoint and OpenOffice.org Impress.



Logos of some graphics, multimedia and presentation software

Specific Purpose Application Software

The software which are designed especially to achieve some specific tasks are called customised software. One of these customised software is the Specific Purpose Application software which is designed keeping in mind the specific requirements of an organisation or an individual.

Examples of this type of software are–reservation system used at airports and railway stations; payroll system used by organisations; stock (inventory control) system used by organisations; billing system used at supermarkets, etc.

Quick Question



Given below is a list of software. Classify them under the appropriate columns.

Backup Utility, Excel, Word, Windows 7, Disk Defragmenter, Reservation System, Stock Control System, Linux

Operating System	General Purpose Application Software	Specific Purpose Application Software	Utility Software



Relationship Between Hardware and Software

The figure given below displays a layered structure which establishes the relationship between various components of a computer system—hardware and software.



Fact Bytes



On-Screen keyboard is a software that lets you type without using the keyboard. It is available under:
Start → All Programs → Accessories → Ease of Access → On-Screen Keyboard.



Let's Summarise



- ▶ The physical components of a computer that we can see and touch are called hardware.
- ▶ Software refers to the computer programs that are needed for running or operating a computer system.
- ▶ There are two broad categories of software—System Software and Application Software.
- ▶ The system software is required to control the operations of a computer system.
- ▶ System software can be classified into three categories—operating system, language processor and utility software.
- ▶ An application software is a set of programs designed to help the user perform specific tasks. It can be classified into general purpose application software and specific purpose application software.



Practice Time

A. Tick (✓) the correct option.

1. _____ is a system software.

a. Presentation ☐

b. Operating system ☐

c. Multimedia ☐

2. _____ is a application software.

a. DBMS ☐

b. Operating System ☐

c. Linux ☐



3. _____ is an example of operating system.

a. UNIX

b. DBMS

c. MICR

4. _____ software is used for creating text based documents.

a. DBMS

b. Spreadsheet

c. Word processing

B. Write 'T' for true and 'F' for false statements.

1. We cannot touch the hardware of the computer.

2. The application software is designed to help the users to perform specific tasks.

3. DBMS is used for creating designs and layout of books.

4. Antivirus software is a type of Utility Software.

C. Fill in the blanks.

1. _____ software is used to create and edit images.

2. _____ software is used to combine images.

3. _____ software is used to design and create slideshows.

4. _____ and _____ are the two types of software.

D. Answer the following questions.

1. Differentiate between hardware and software.

2. What is operating system? What is its use.

3. Give details of any two utility software.

4. Differentiate between general and specific purpose application software.

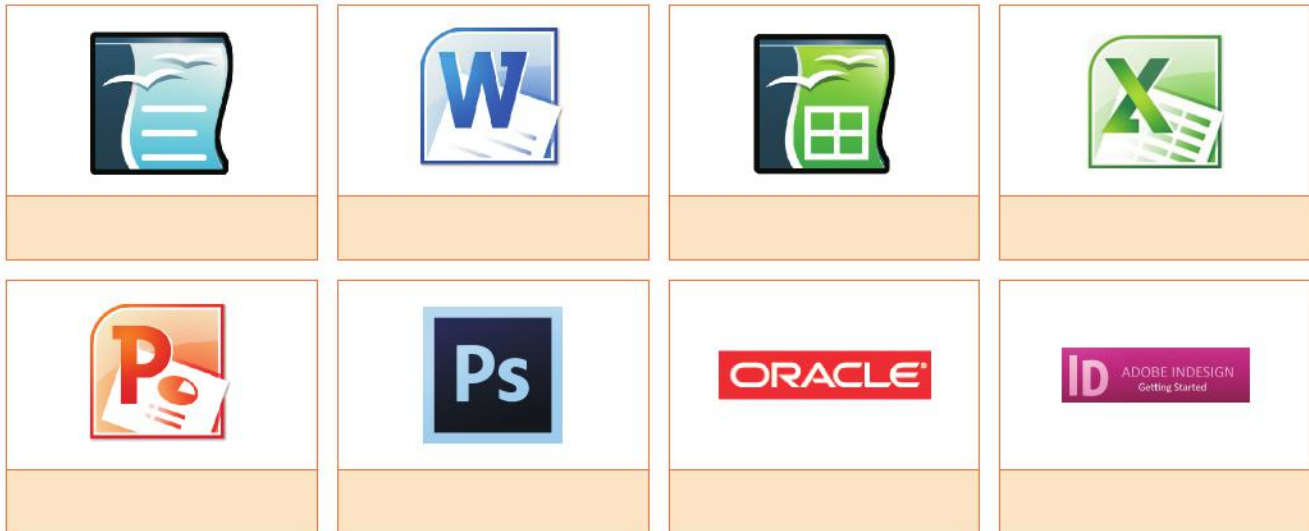
5. Differentiate between system software and application software.



Fun Time



- Identify the following softwares and name them.



Lab Time



- Visit the computer lab. Find out the operating system and the antivirus software being used in your school.
- Prepare a list of software installed in your computer system at home. Classify them as system software or application software. Further, classify the application software as general purpose application software and specific purpose application software.
- Make use of antivirus software to scan the computer system for viruses.

Brain Booster



- Computer without software is a dumb machine. Why?

Project Time



- Prepare a presentation on 'Computer Software'.

Teacher's Note : The teachers can explain the concept of software by giving suitable examples of the software used by students on a computer to perform various tasks.



Personalising Windows Environment

3.



This chapter includes



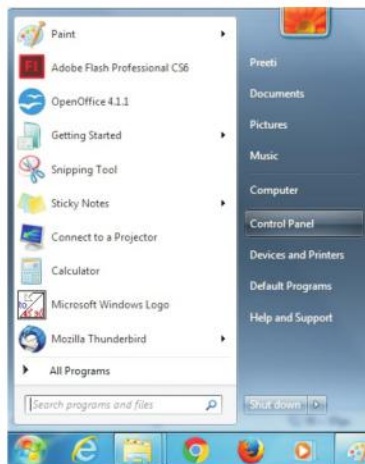
- ▶ Changing the Display Settings
- ▶ Changing the Screen Resolution
- ▶ Changing Date and Time Settings

As we have already studied about the various types of software. One of them is the Operating System which is the most basic software.

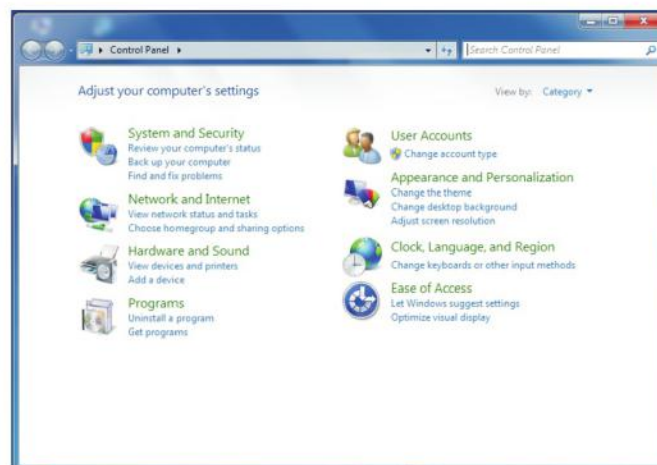
The Operating System is the software which acts as an interface between the user and the hardware.

Microsoft Windows is the most popular OS. The various versions of Windows OS available today include Windows 7, Window 8 and Windows 10. When working with Windows, you can customise the appearance and functionality of the computer according to your preferences. This task of customising the computer can be done using the **Control Panel**.

The **Control Panel** window can be opened from the Start menu.



Opening Control Panel



Control Panel window

In this chapter, we will learn about changing display and date/time settings in Windows 7 through **Control Panel**.

Changing the Display Settings

You can personalise various display settings such as the background picture of the desktop or appearance of windows elements on your computer using **Themes**. Also,



you can change the screen resolution for better clarity of the text and images displayed on a screen. These options can be accessed using the **Appearance and Personalization** section in the Control Panel window.

Let us discuss more about these display settings.

Changing Themes

A **theme** is a combination of pictures, colours and sound on the computer.

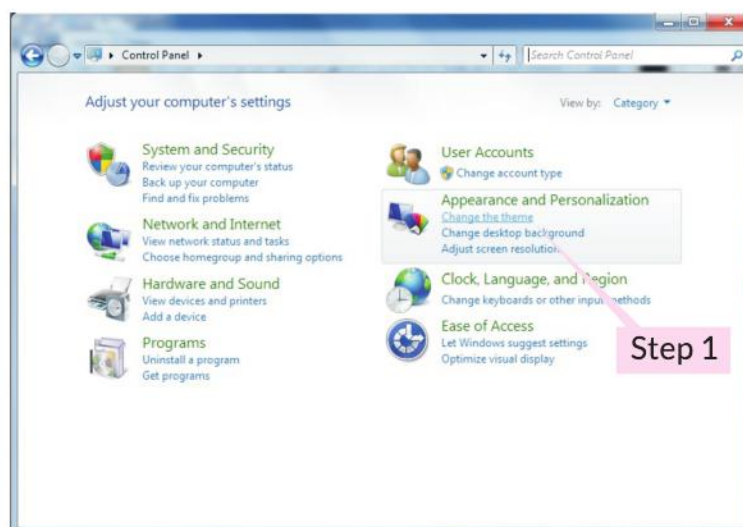
It includes a desktop background, window border colour, a sound scheme and a screen saver.

A **screen saver** is any moving picture or graphic that is displayed on the screen when you do not use the mouse or the keyboard for some specific time.

Windows comes with various preinstalled themes. These include **Aero Themes**, **Windows 7 Basic** theme and **High Contrast** themes. To apply themes, follow the given steps:

1. Click on the **Change the theme** link under the **Appearance and Personalization** section in the **Control Panel** window. This displays the **Personalization** window.
2. In the **Personalization** window, click on the desired theme to apply it. When you apply a theme, the desktop picture, colour of window frames and screen saver of the selected theme are applied. In this example, we have chosen the **Nature** theme under **Aero Themes**.

You can change individual elements of the theme—desktop background, screen saver and window colour as per your own preferences.



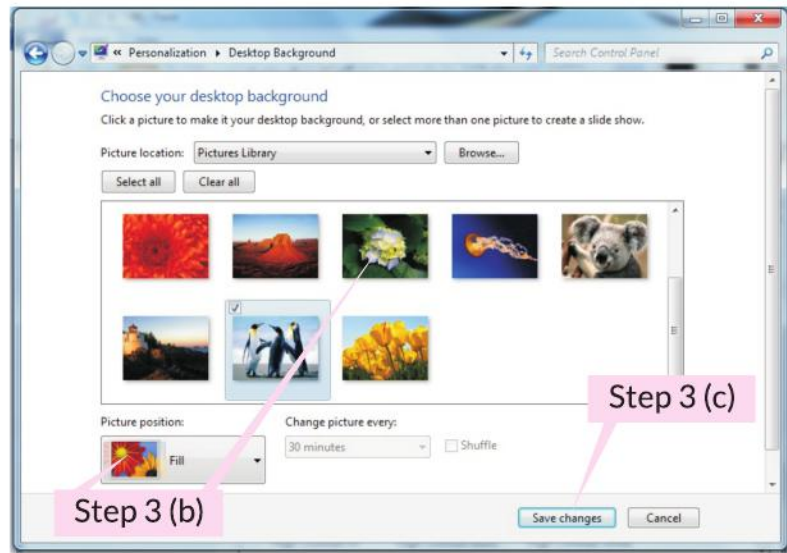
Control Panel window



Personalization window

3. To change the desktop background, follow the given steps:

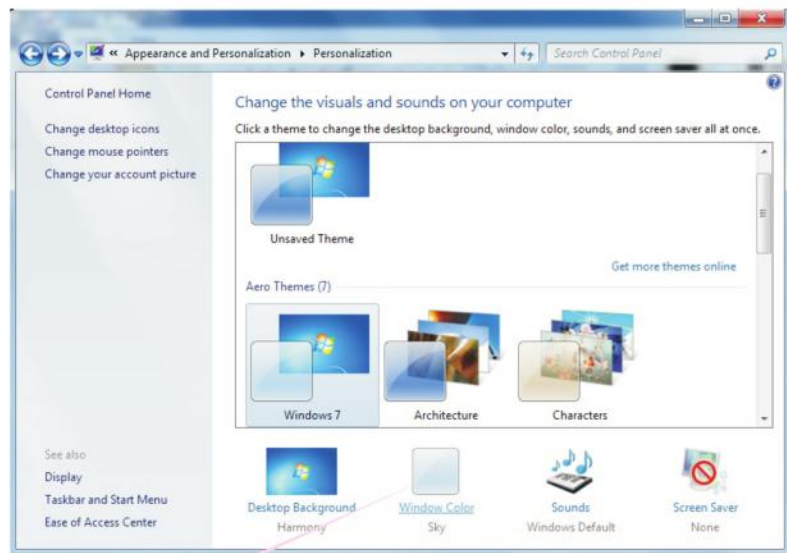
- (a) Click on the **Desktop Background** link in the **Personalization** window. It displays a thumbnail view of the pictures available under the selected theme.
- (b) Click on a picture to apply it as the desktop background. You can also change how the background picture should be displayed using the **Picture position** drop-down box .
- (c) Click on the **Save changes** button to apply the settings.



Desktop Background window

4. To change the window colour, follow the given steps:

- (a) Click on the **Window Color** link in the **Personalization** window. A list of colours available is displayed .
- (b) Click on the colour you want.
- (c) Click on the **Save changes** button to apply the settings.



Personalization window

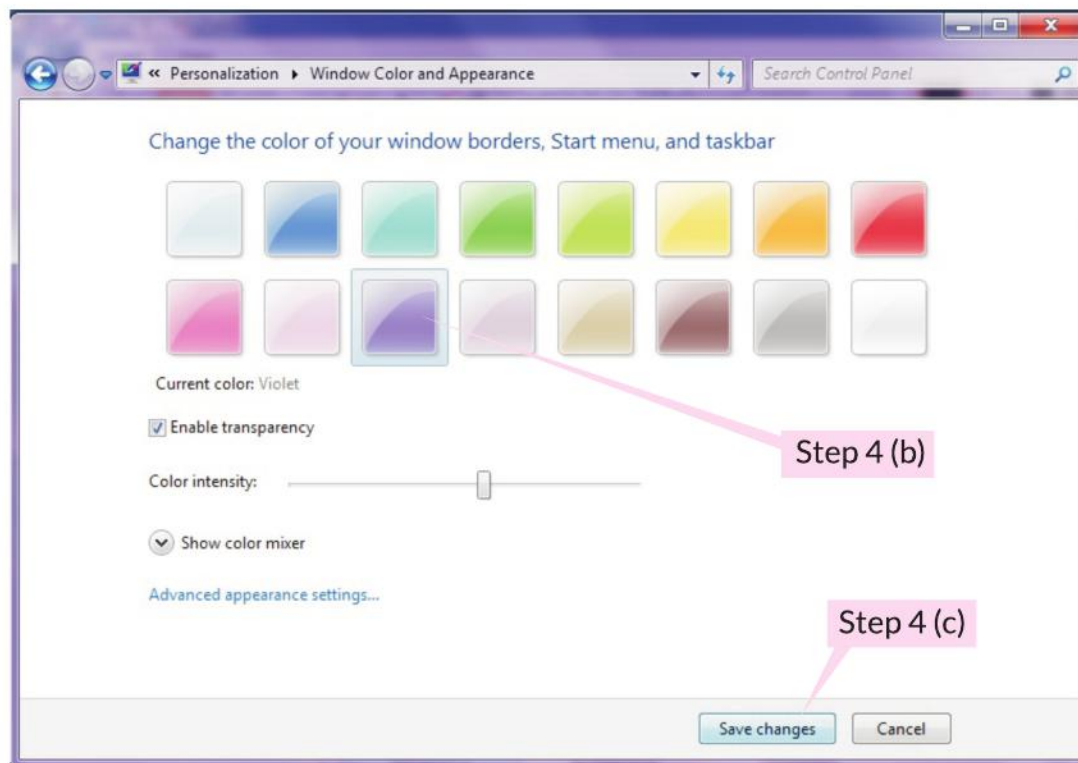
On Your Tips



The option to open the **Personalization** window can also be accessed by right-clicking on the desktop and choosing the **Personalize** option from the shortcut menu.

You can also create desktop background slide show by selecting multiple pictures. For this, select the check box for each picture and specify the timings after which the next picture should be displayed in the **Change picture every** box in the **Desktop Background** window.



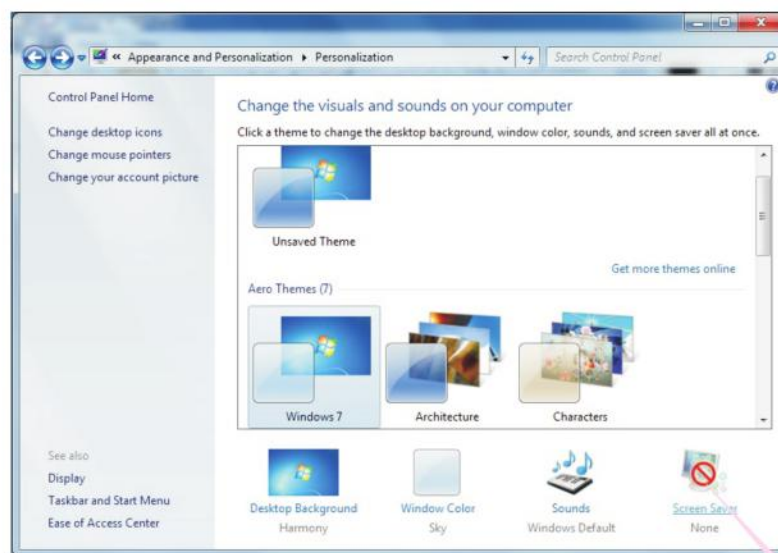


Window Color and Appearance window

5. To change the screen saver, follow the given steps:

(a) Click on the **Screen Saver** link in the **Personalization** window. This displays the **Screen Saver Settings** dialog box.

(b) In the **Screen Saver Settings** dialog box, do the following:

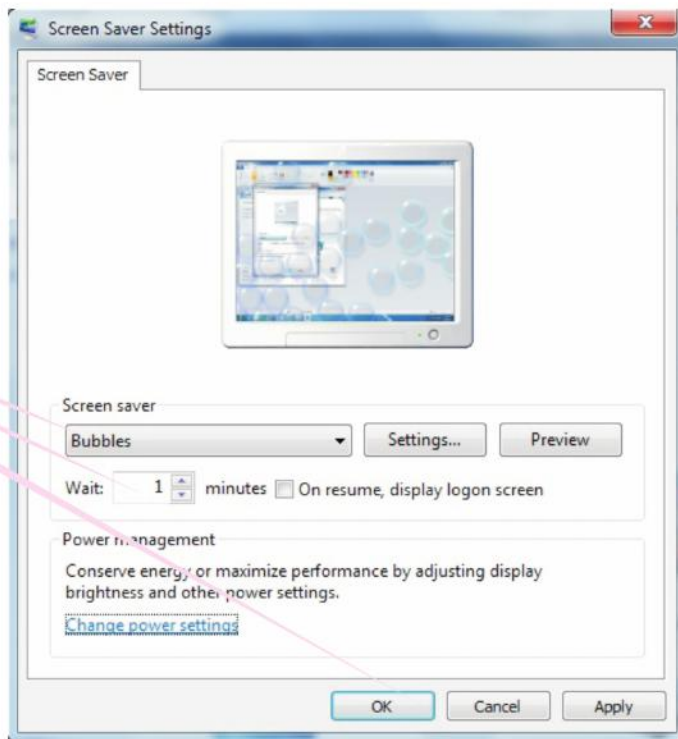


Personalization window

Step 5 (a)

- Choose the screen saver that you want to use from the screen saver list.
- In the **Wait** box, specify the minutes for which the computer should remain idle before the screen saver is displayed.
- Click on **OK**.





Screen Saver Settings dialog box

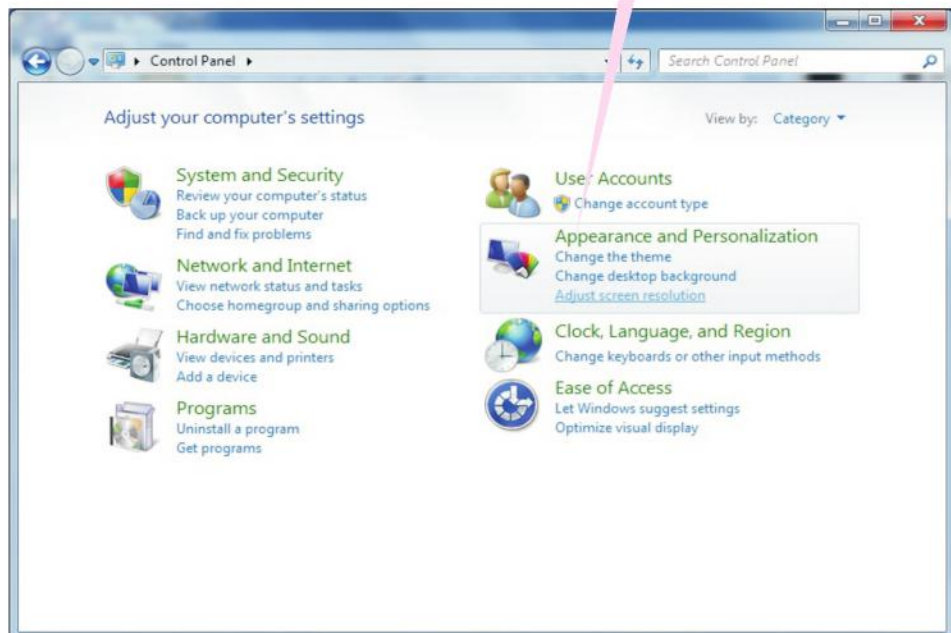
On Your Tips



You can also display your pictures as screen saver by choosing the **Photos** option from the Screen saver drop-down box. By default, it displays the pictures in the Pictures folder as a slide show. To specify a different location of the folder containing pictures and other options such as picture size, click on the **Settings** button.

Changing the Screen Resolution

It is due to the screen resolution that determines the amount of information that you can view on your computer screen at any given time. The higher screen Resolution makes everything appear small and the low screen resolution helps to make everything appear large. Follow the given steps to change the screen resolution:

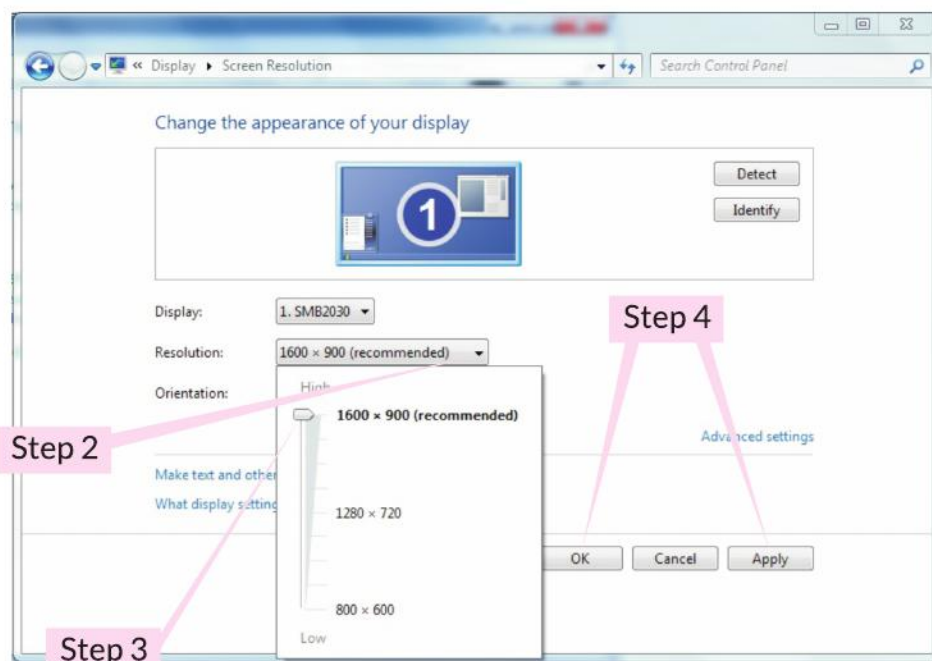


Control Panel window

1. Click on the **Adjust screen resolution** link under the **Appearance and Personalization** section in the **Control Panel** window. This displays the **Screen Resolution** window .

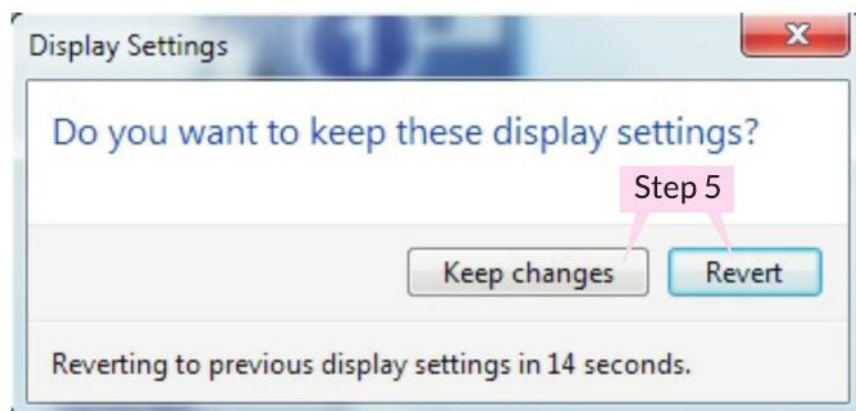


2. Click on the drop-down arrow next to **Resolution**.
3. Move the slider to the resolution you want.
4. Click on the **OK** or **Apply** button. The screen resolution changes accordingly and a **Display Settings** dialog box appears.



Screen Resolution window

5. Click on **Keep changes** to use the new resolution or click **Revert** to go back to the previous resolution.

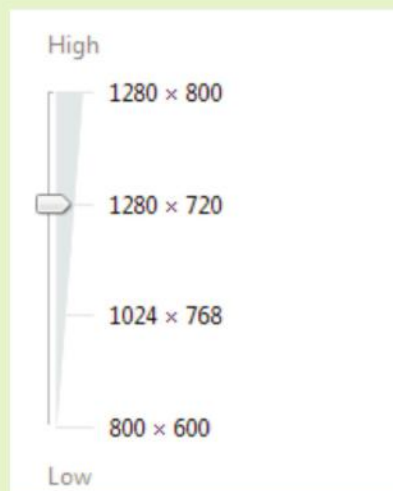


Display Settings dialog box

Fact Bytes



LCD Monitors, including laptop screens, typically run best at their native (recommended) resolution. It is recommended to use your monitor's native resolution so that you see the sharpest text and images. You will notice the word "recommended" next to the ideal resolution setting.



Quick Question



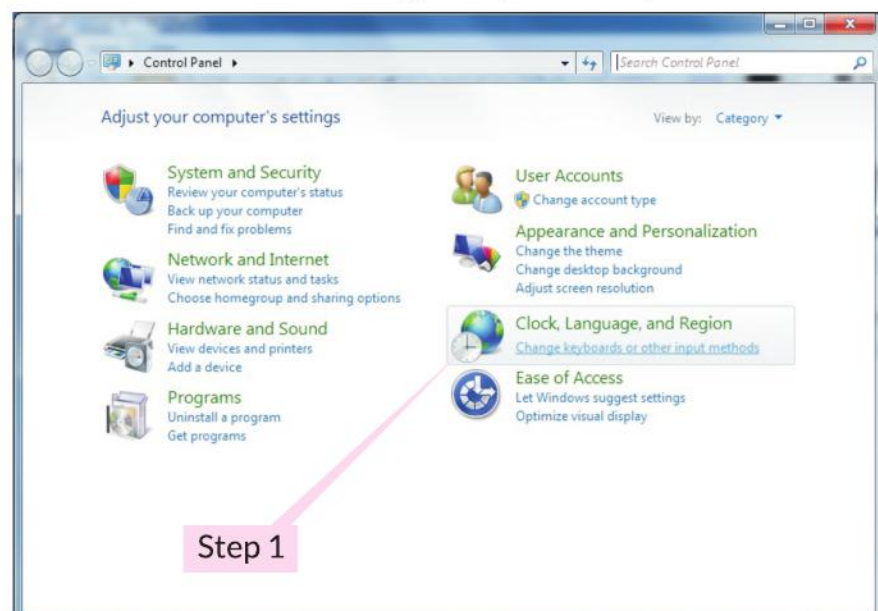
Match the following terms with their descriptions.

Terms	Description
(a) Screen saver	(i) A combination of pictures, colours and sound on your computer.
(b) Screen resolution	(ii) A window that provides options for customising the appearance and functionality of the computer according to you preferences.
(c) Theme	(iii) Any moving picture or graphic that is displayed on the screen when you do not use the mouse or the keyboard for some specific time.
(d) Control Panel	(iv) The amount of information you can view on the computer screen at any given time.

Changing Date and Time Settings

Follow the given steps to change the date and time settings on your computer.

1. Click on the **Clock, Language, and Region** link in the **Control Panel** window. A set of options related to '**Date and Time**' and '**Region and Language**' appear in the window.
2. Click on the **Set the time and date** link. The **Date and Time** dialog box appears.
3. Click on the **Change date and time** button under the **Date and Time** tab. The **Date and Time Settings** dialog box appears.



Control Panel window

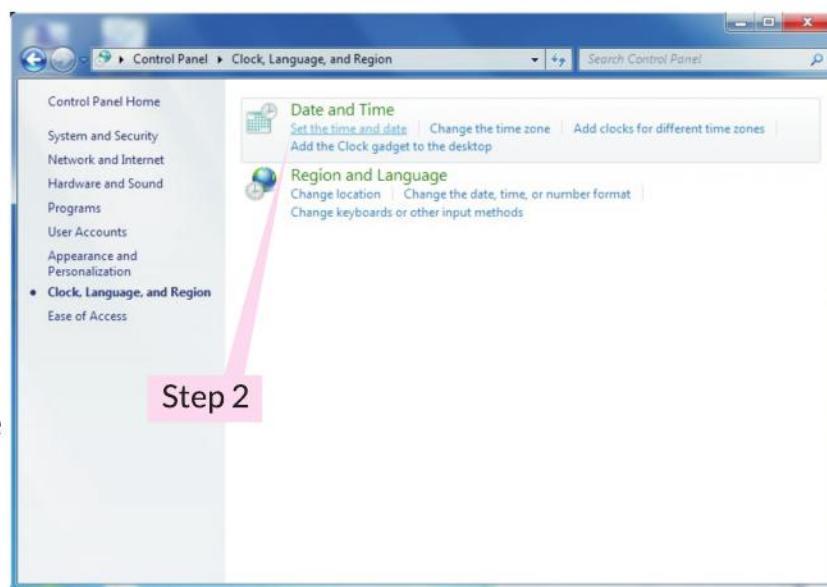


4. In the **Date and Time Settings** dialog box, do one or more of the following:

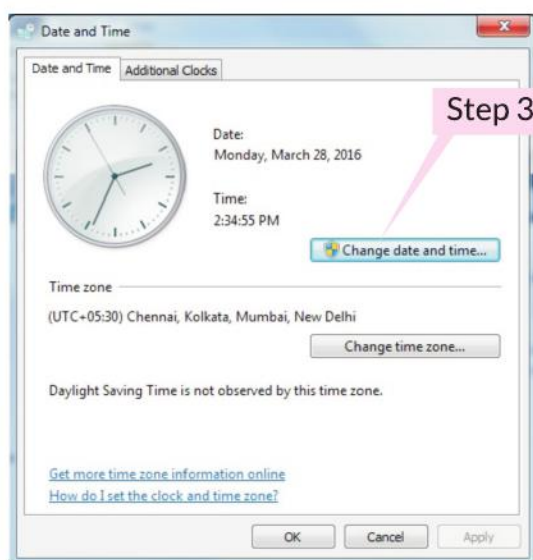
(a) To change the date, click on the black left and right arrows to move to the correct month and year. Then select the date you want.

(b) To change the time, click on the up and down arrows next to the clock or click the **hour, minutes or seconds** box and type the appropriate value.

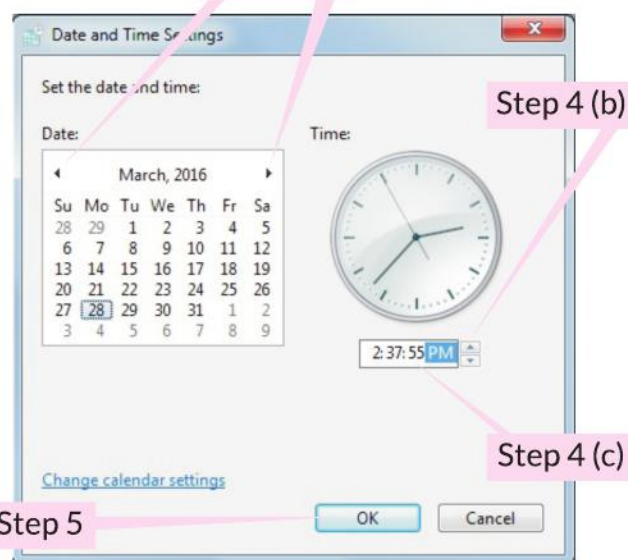
(c) To indicate **AM** or **PM** for the time value, select it and then click on the up arrow to the down arrow to switch between the two values.



5. Click on **OK** in the **Date and Time Settings** dialog box and **Date and Time** dialog box when the date and time are correct.



Date and Time dialog box



Date and Time Settings dialog box

On Your Tips



A quicker way to change the date and time settings is to click on the date/time in the **Notification Area** on the **Taskbar**.

Let's Summarise



- ▶ The operating system acts as an interface between the user and the hardware.
- ▶ Microsoft Windows is the most popular operating system.
- ▶ You can customise the appearance and functionality of the computer according to your preferences using the Control Panel.
- ▶ A theme is a combination of pictures, colours and sound on the computer.
- ▶ A screen saver is any moving picture or graphic that is displayed on the screen when you do not use the mouse or the keyboard for some specific time.
- ▶ The screen resolution determines the amount of information you can view on the computer screen at any given time.
- ▶ The options to change the theme and adjust the screen resolution are available under the Appearance and Personalization section in the Control Panel window.
- ▶ The option to change the Date and Time Settings can be accessed by choosing the Clock, Language, and Region link in the Control Panel window.



Practice Time

A. Tick (✓) the correct option.

1. _____ is a moving graphic or screen that appears when mouse or keyboard is not in use.
(a) Background ☐ (b) Screen saver ☐ (c) Windows ☐
2. _____ is a combination of pictures, colours and sounds on a computer.
(a) Theme ☐ (b) Screen Resolution ☐ (c) Screen saver ☐
3. _____ is the most basic software.
(a) Operating System ☐ (b) Utility Software ☐ (c) Graphics ☐
4. The customising of computer can be done using _____.
(a) Editing ☐ (b) Control Panel ☐ (c) Toolbar ☐

B. Write 'T' for true and 'F' for false statements.

1. Microsoft windows is the most popular operating system.
2. The personalization of themes is not possible.
3. Aero Theme is a type of preinstalled theme.
4. Only once you can change the Screensaver.



C. Fill in the blanks.

1. _____ resolution makes everything on the screen large.
2. Appearance & personalisation section is present in _____ window.
3. Aero is a type of pre installed _____.
4. The quicker way to change date and time is to click date and time in the _____ on the Taskbar.
5. The _____ determines the amount of information to be viewed on computer screen.

D. Answer the following questions.

1. What is the significance of the Control Panel window? How can you open it?

2. Define theme. Name some preinstalled themes.

3. Write the steps of changing the screen saver.

4. Write the steps of changing a theme.

5. Write the steps of changing the date and time settings of your computer.

6. What do you understand by screen resolution? How does it affect the display of items on the screen?

Lab Time



► **Visit your computer lab and perform these tasks.**

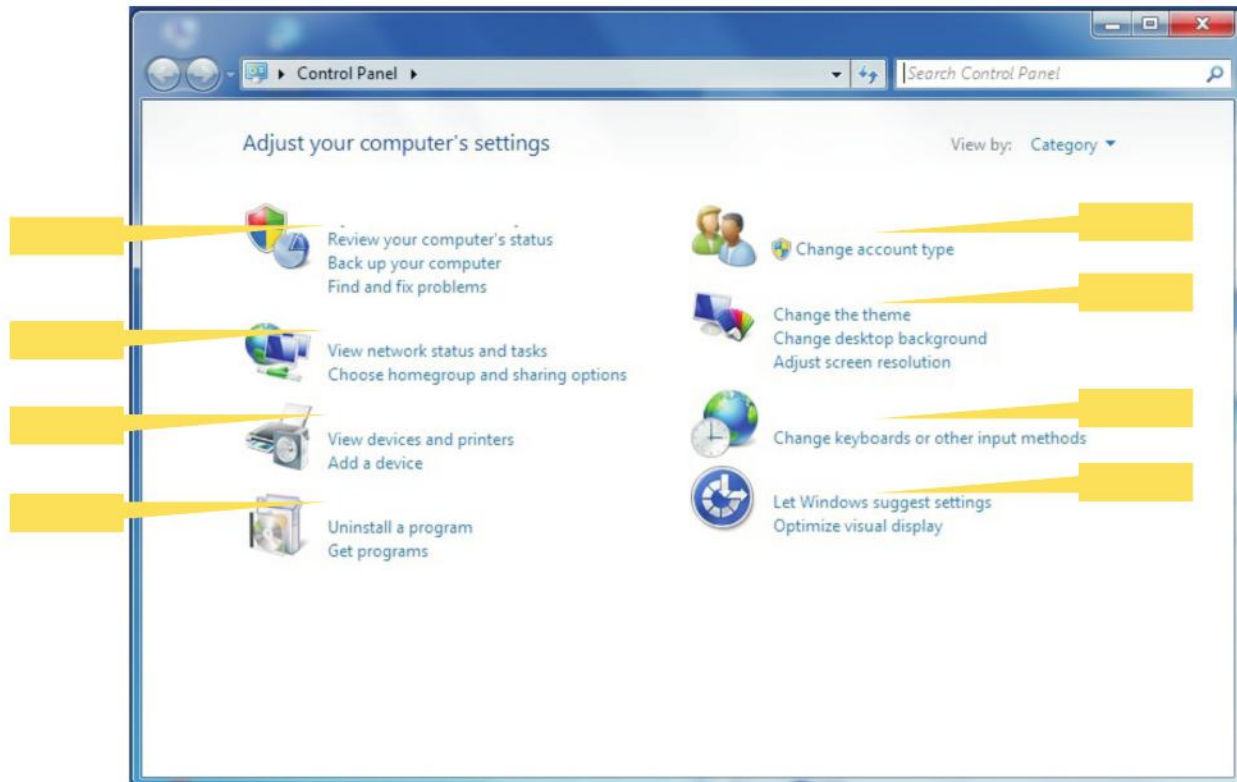
- (a) Apply the Aero theme.
- (b) Change the desktop background of the applied theme.
- (c) Set or change the screen saver of the applied theme.
- (d) Change the colour of the window borders.
- (e) Decrease the screen resolution.
- (f) Update the date and time of the computer to match the current date and time.



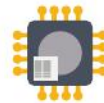
Fun Time



- Label the figure given below.



Brain Booster



- If you want to change the date and time of your computer without using Control Panel, then what will you do?

Project Time



- Prepare a presentation on– 'Personalising Windows Environment'.

Teacher's Note : The teacher can tell the students about the various versions of Windows Operating System available today and also ask them to customise the appearance and functionality of their computer.



Advanced Features of MS Word

4.



This chapter includes



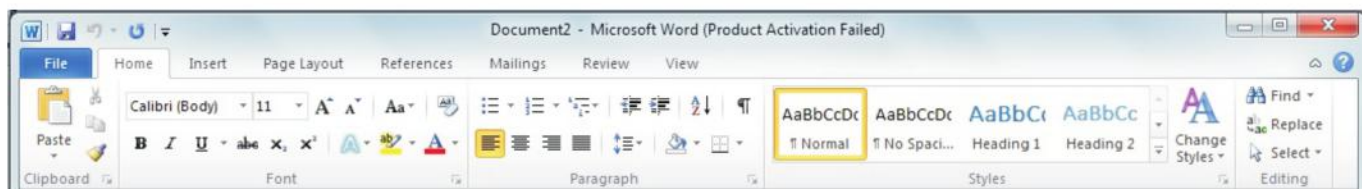
- ▶ Character Formatting
- ▶ Paragraph Formatting
- ▶ Borders and Shading
- ▶ Inserting Clip Art
- ▶ Inserting Predesigned Shapes
- ▶ Text Formatting using Font Dialog Box
- ▶ Creating Lists
- ▶ WordArt
- ▶ Inserting Stored Images

In MS Word, formatting is the general arrangement or layout of the text you type. Formatting the text makes it look neat and easy to read. The way you format your text in a document depends on how you intend to use it and how you want it to look.

Character Formatting

Text is the major constituent of our document. When we enter the text in the Word document, it appears with some standard formatting. We can always change these standard formattings as per the need of our document. Character formatting deals with the representation of text with a suitable style.

Microsoft Word 2010 has many such features which can be accessed from the **Font** group on the **Home** tab.



Fact Bytes



Formatting is a general arrangement of the text to make the document attractive and presentable.



Changing Font Type

The style that is used for typing in the document i.e. the character formulations is the meaning of **Font**. Different font types are recognized by their unique names. The default font type used for typing in Word document is 'Calibri' font.

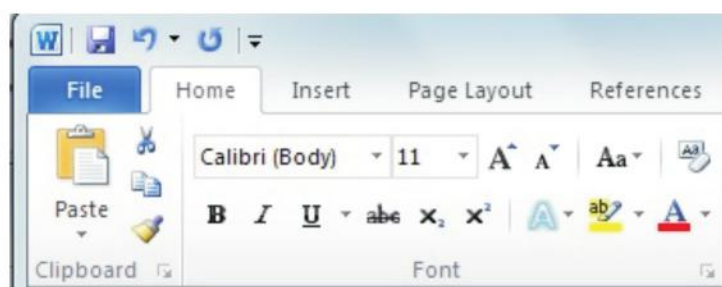
- ▶ Select the text to change its default font type
- ▶ Click the **Font List**  drop box in the **Font** group on the **Home** tab.
- ▶ The list shows the various fonts. Click the desired font style to apply on the selected text.



Font Style

To highlight the specific text like some heading or title, special font effects can be applied on the default regular style text.

- ▶ Select the text.
 - ▶ Apply the style from the following options in **font** group.
- Bold (B)** : To add weight to the text.
- Italic (I)** : To show the text in tilted style.
- Underline (U)** : To show a line under the selected text.



On Your Tips



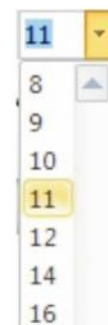
You can press **Ctrl + B** to Bold, **Ctrl + I** to make it Italic and **Ctrl + U** to Underline the selected text.

Font Size

The normal text in word document appears in 11 font size.

To increase or decrease the character size:

- ▶ Select the text.
- ▶ Click Size  drop box to choose the new font size in number.



Font Colour

In MS Word, the default colour of the text is black. But we can change the colour of font to any other suitable one like this:

- Select the text.
- Click **Font Colour** () drop box in the **Font** group and choose the desired colour.

Text Highlighting

This formatting option works like a highlighter pen which is used for highlighting the text with a background colour without changing its text format.

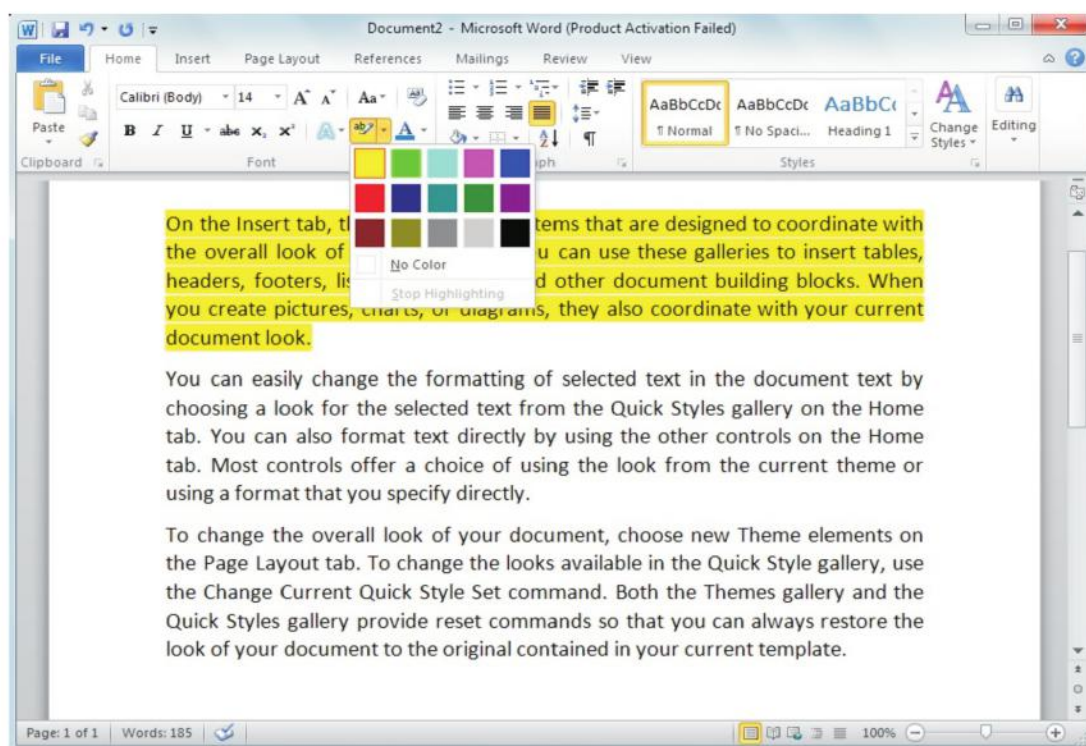
- Select the text to be highlighted.
- Click the **Text Highlight Colour**  drop button in the **Font** group to choose the pen colour. The selected text will appear in highlighted mode.

If you want to highlight the text at multiple locations, click the **Text Highlight** button to select the pen colour. Now one by one select those text blocks which are to be highlighted.

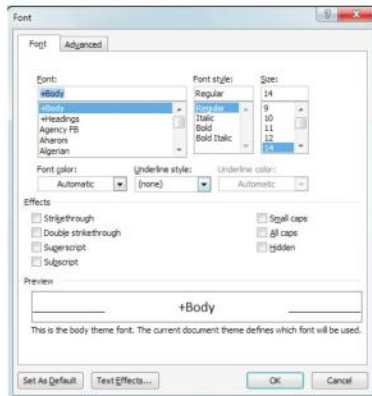
On Your Tips



You can also select the highlight colour first using highlight button. Your mouse pointer will change to a pen shape. Select the text with this pointer to highlight the same with your selected colour. Press ESC key to Turn Off the Highlight option.

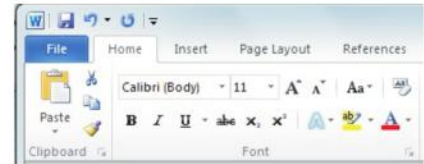


Text Formatting using Font Dialog Box



If you want to change the settings related to font, it can be done using single dialog box.

- Select the text to be formatted.
- Click the **dialog box launcher**  in the lower right corner of **Font** group.
- Font dialog box will appear as shown in figure.
- Use various font settings in it as needed and click **OK** button.



Fact Bytes



The default font used in Microsoft Office 2010 is Calibri and its size is 11 points.

Paragraph Formatting

Paragraph is the text which appears between the two key strokes of enter key. There are formatting features which help to format paragraph text collectively, like text alignments, line spacing, indentations, etc.

Aligning the Text

Alignment refers to the placement of paragraph text between the left and right margins. Word 2010 offers four type of alignments.

Alignment	Example	Text Placement
Left alignment ()	An electronic device for storing and processing data, typically in binary from, according to instructions given to it in a variable program.	Aligns text with respect to the left margin.
Right alignment ()	An electronic device for storing and processing data, typically in binary from, according to instructions given to it in a variable program.	Aligns text with respect to the right margin.
Center alignment ()	An electronic device for storing and processing data, typically in binary from, according to instructions given to it in a variable program.	Aligns the text between left and right margin.
Justify alignment ()	An electronic device for storing and processing data, typically in binary from, according to instructions given to it in a variable program.	Distributes text evenly between left and right margins.



For aligning the specific paragraph(s):

- Select the paragraph (s).
- Click the required alignment option in the **Paragraph** group on the **Home** tab.
- The text or object gets aligned accordingly.



Alignment Buttons

On Your Tips



Keyboard shortcuts: Press **Ctrl + E** to center align, **Ctrl + J** to justify, **Ctrl + L** to left align and **Ctrl + R** to right align.

Fact Bytes



Default text alignment in Microsoft Word is **Left** alignment.

Line Spacing

The space between the two consecutive lines of a paragraph is called **Line Spacing**. By default the paragraph lines appear with single line spacing.

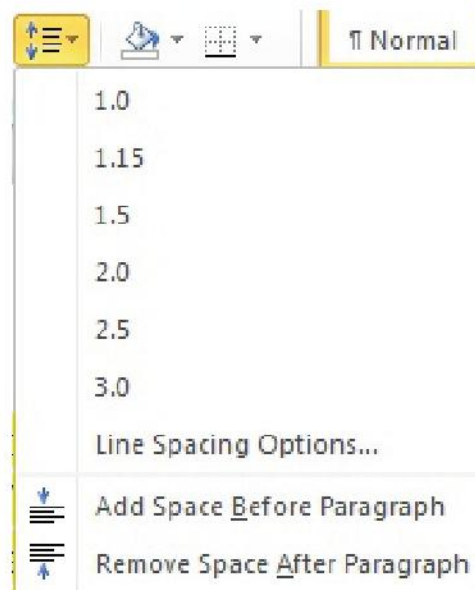
To change this spacing:

- Select the paragraph(s).
- Click **Line Spacing** (↕) option in the **Paragraph** group on the **Home** tab to expand the list.
- Select the required spacing from the list to apply the changes.

Fact Bytes



The Default line spacing in Microsoft word 1.15, i.e., single line spacing.



Indenting Text

It is also a formatting feature which is used for moving the paragraph away from left or right margins to draw readers attention.

- Select the paragraph(s) to be indented.
- Click **Increase Indent** (↗) option in the **Paragraph** group.
- The selected paragraph(s) will shift towards right by 0.5 inch. Everytime you click the increase indent button, the text moves by 0.5 inch towards right.
- Click **Decrease Indent** (↖) option to reduce the indentation or to move back towards the left by 0.5 inch with every click.



Creating Lists

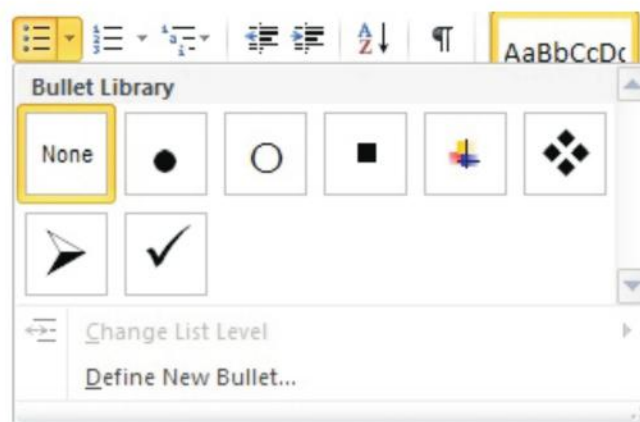
Word 2010 offers facility to create lists automatically. Two types of list can be created in the document; numbered (ordered) and bulleted (unordered) list.

Bulleted list

This type of list is used, where order of the list items is not important, the list items are marked with a selected symbol.

To create such type of list:

- ▶ Click in the document to select the list position.
- ▶ Click **Bullets** (≡ ▸) drop option to open bullets library.
- ▶ Click and select the suitable symbol to be used. The symbol will appear at the insertion point.
- ▶ Enter list item. Press **Enter** key to add next item.

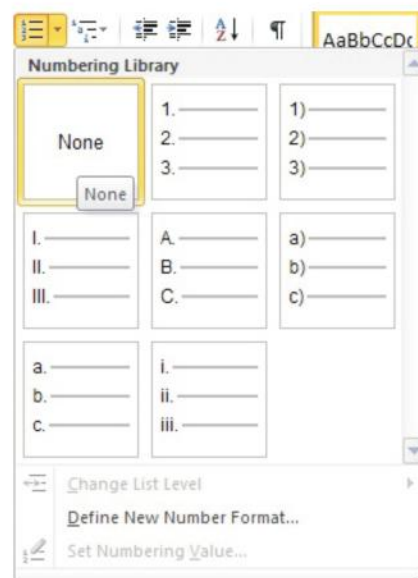


Numbered list

Numbered or ordered list is used to mark the list items with a unique number or alphabet. These numbers appear automatically in the increasing order to add list items.

To create this type of list:

- ▶ Click in the document from where the list is to be started.
- ▶ Click **Numbering** (≡ ▸) drop option in **Paragraph** group to open Numbering library.
- ▶ Select the numbering style.
- ▶ The first number of the list will appear. Enter the list item and press **Enter** key to get the next number.



Fact Bytes

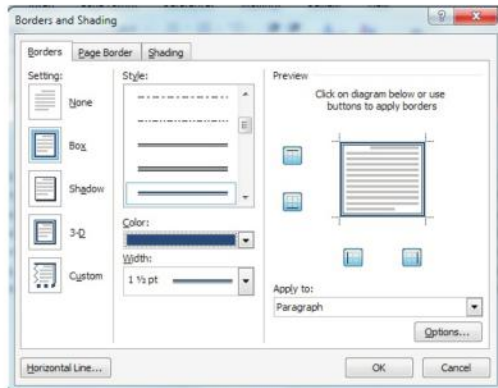


- By default Microsoft Word applies black dots as the bullets to your list.
- Roman letters numbering style is used by default to make an ordered list.



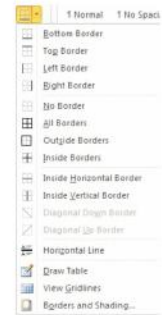
Borders and Shading

To further enhance the text appearance, we can also add text/paragraph or page border as needed. A background colour added gives shading effects which further adds up to its enhancement.

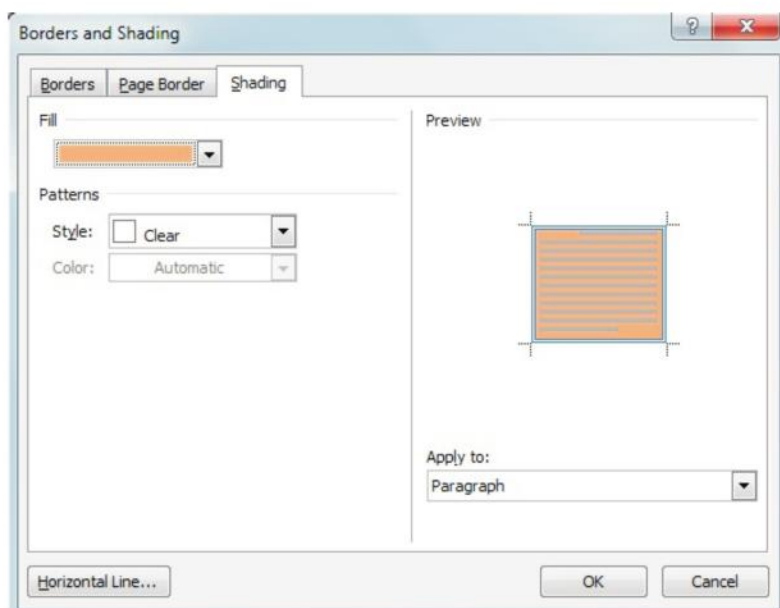
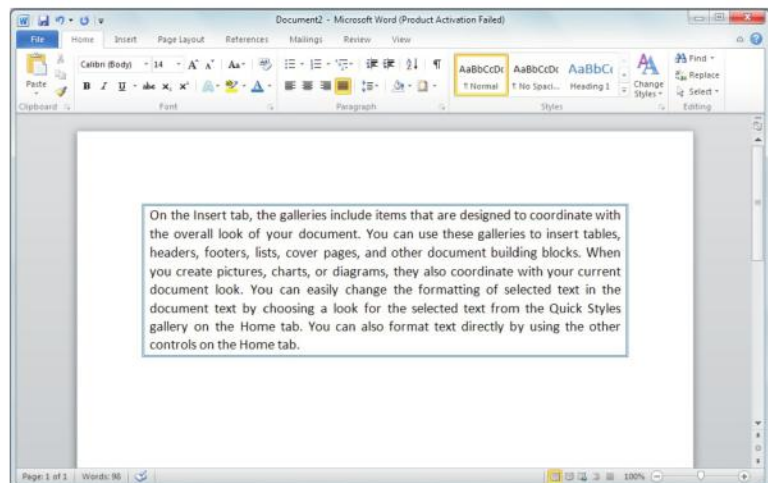


To Add Text Border

- Select the text/paragraph to be bordered.
- Click **Border** () drop button in the **Paragraph** group.
- Choose **Borders and Shading** option to open the dialog box as shown in figure. Click **Borders** tab in it.



- Choose border setting as Box / Shadow / 3D.
- Choose border line style and its colour. Change the border thickness if needed.
- Click on the **OK** button.

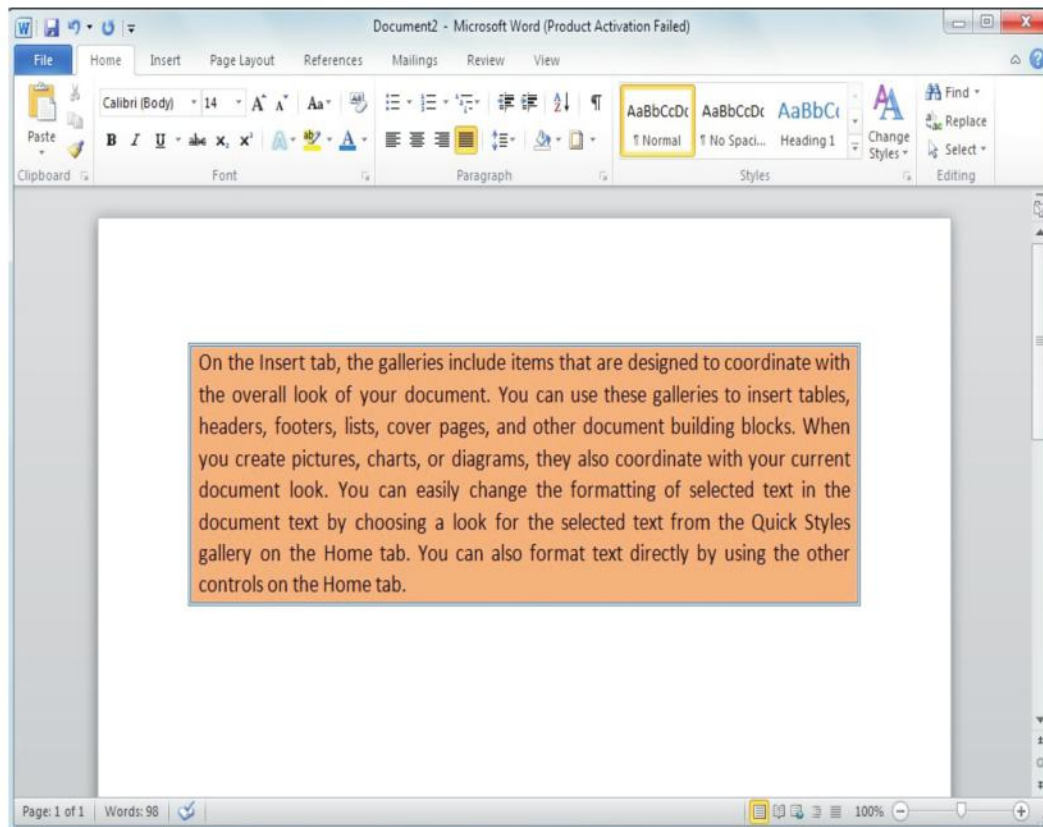


Adding Background Shades

To apply the background shade:

- Click **Shading** tab in the **Borders and Shading** dialog box.
- Choose the background fill colour.
- Select a pattern, if needed.
- Click on the **OK** button.



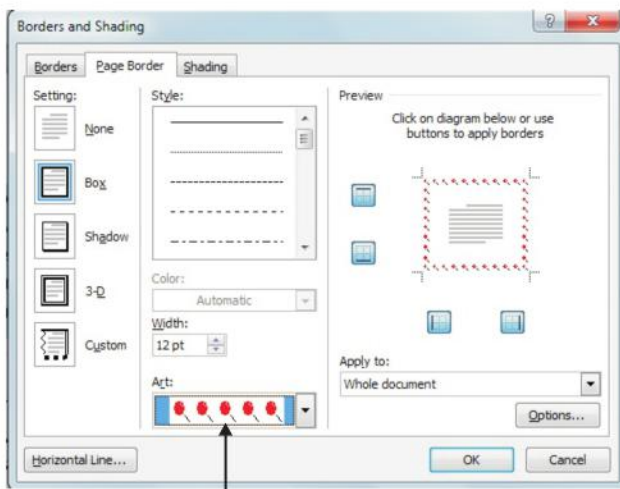


Applying Page Border

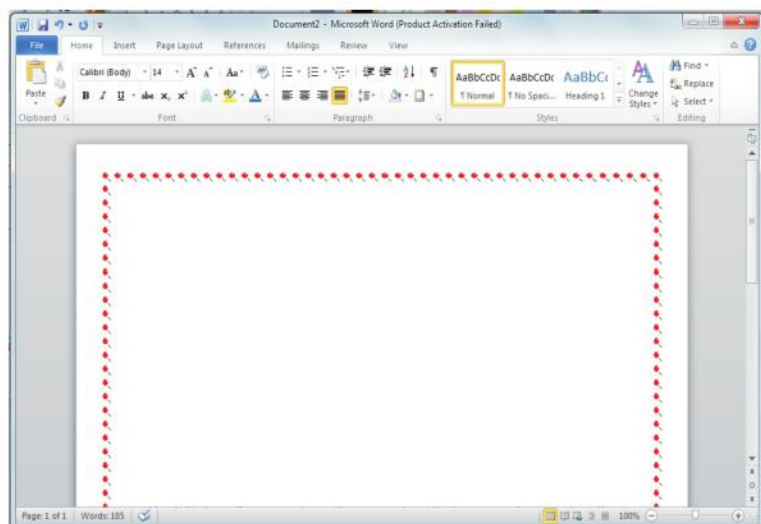
Borders can also be added to enclose the entire page contents.

To add Page Border:

- Click **Page Border** tab in the border and shading dialog box.
- Choose the border settings as needed. You can also select an artistic border for the page.
- Click on the **OK** button.



Click here for artistic border



Word Art

WordArt is a gallery of text styles which helps to add special text effects in the document like changing the shape of inserted text, adding gradient effects, etc. WordArt is added in a text placeholder, which can be moved or repositioned anywhere on the document.

Adding WordArt

To add WordArt Text:

- Click in the document to select the position where WordArt is to be inserted.



- Select **Insert** tab and click WordArt option in the **Text** group to open Word Art gallery.
- Select Word Art style. The WordArt textbox will appear at the selected location.
- Type the desired text. Click outside the text box to place the text.

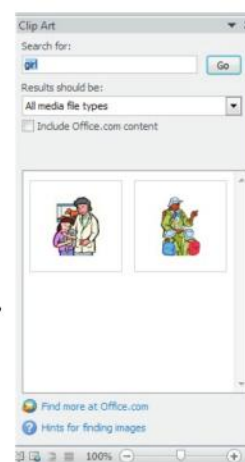


Inserting Clip Art

Clip Art is a built in collection of readymade images provided with the MS Office components. We can use these Clip Arts in our document to illustrate its contents.

To insert the Clip Art:

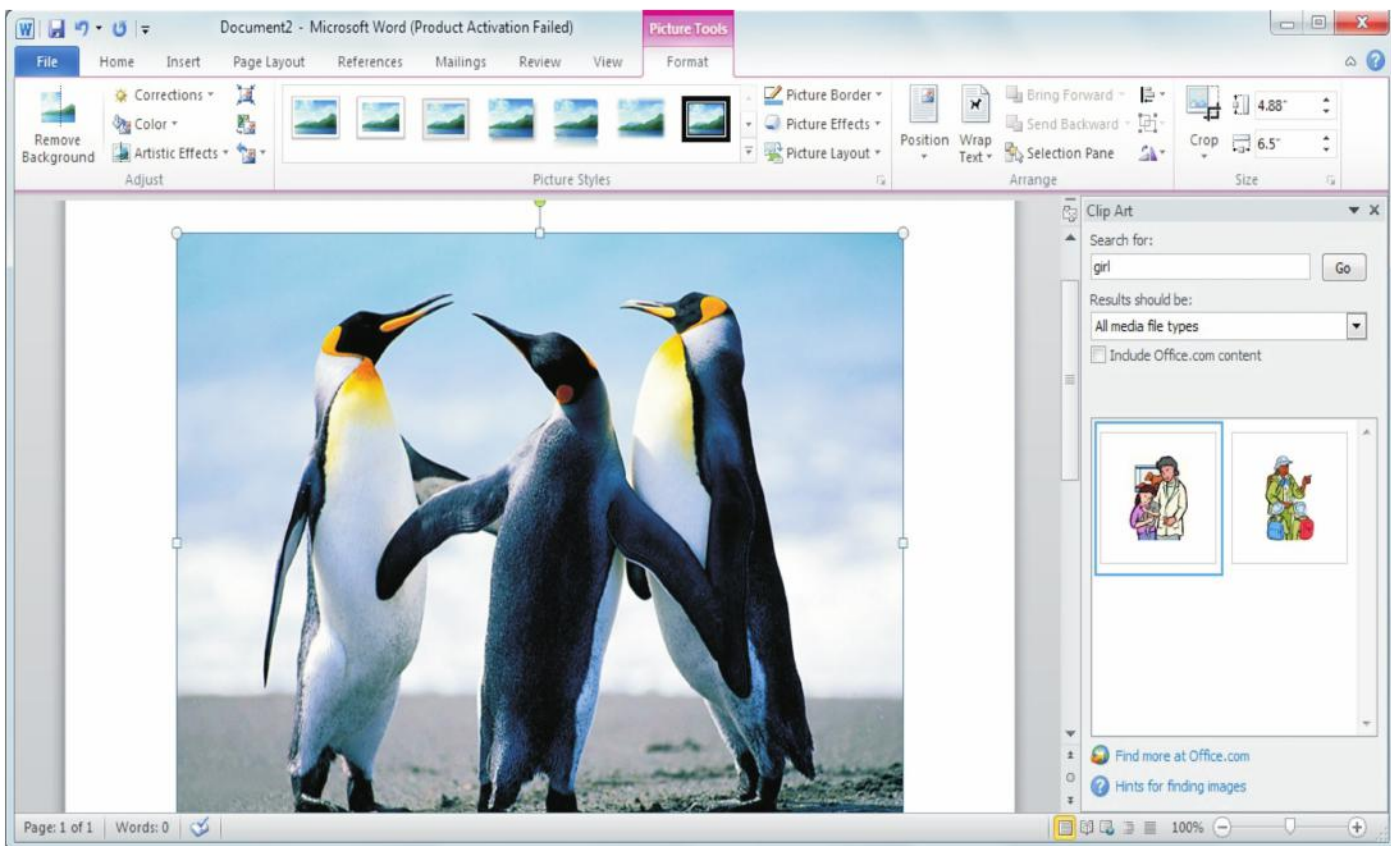
- Click and select the position in the document.
- Select **Insert** tab. In the **Illustrations** group click on **Clip Art** option.
- ClipArt taskpane will appear as shown in figure.
- Enter the word or phrase that describes your illustration in the **Search for** box.
- Click **Go** button.
- A list of related images will appear.
- Click the image in the list you want to add.
- The image will appear at the insertion point.



Inserting Stored Images

You can add pictures or images in the word document. The pictures or images stored in our computer can also be used in the Word document.

- Click where you want to insert the picture.
- On the **Insert** tab in the **Illustrations** group, click **Picture**  option.
- **Insert Picture** dialog box will appear as shown in the figure.
- Locate the picture in the storage device. Click the image to select it and use **Insert** button to add it in your document.

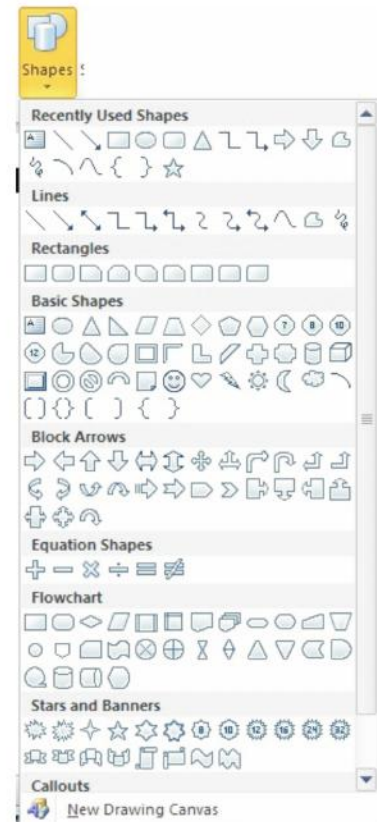
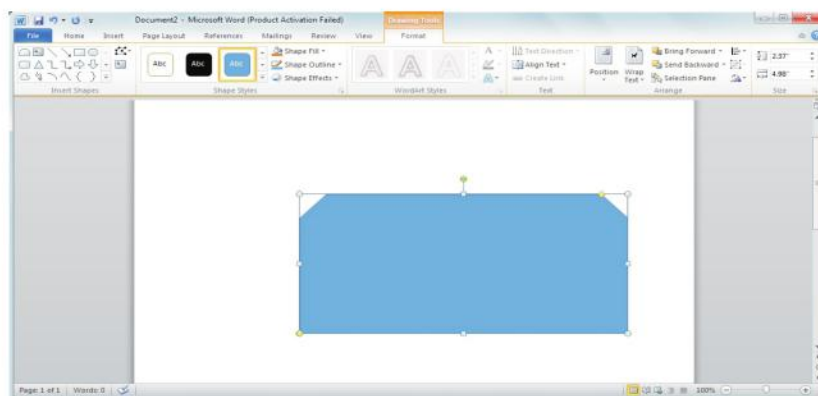


Inserting Predesigned Shapes

Word offers a wide range of predesigned shapes like lines, basic shapes, callouts, etc. to draw them quickly and easily in the document.

To draw any of these shapes;

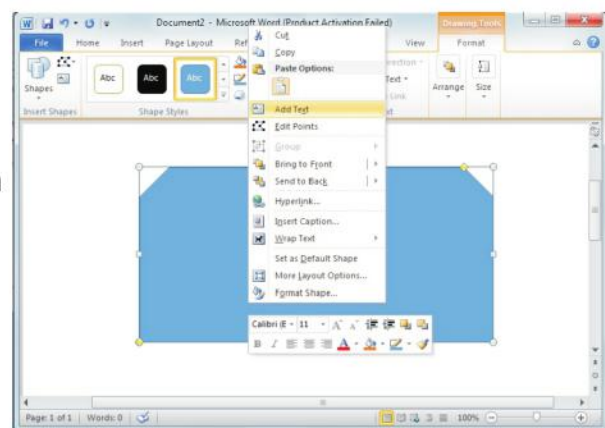
- Click on the **Insert** tab.
- Click Shapes  drop button in the **Illustrations** groups.
- Choose the required shape to be drawn.
- Click and drag in the document area to draw the figure.



Adding Text to the Shape

If you want to enter some text in the inserted image.

- Click and select the shape.
- Right click on it and choose **Add Text** option in the shortcut menu.
- Insertion point will appear in the shape which can be used for typing text in it.



Fact Bytes



Shapes is a gallery with many types of predesigned shapes which can be quickly drawn in the document like lines, geometrical shapes, callouts, banners, stars, etc.

Quick Question



Name five shapes that can be drawn using shapes tool.



Let's Summarise



- ▶ Formatting means the general arrangement of document contents to make it more attractive and readable.
- ▶ Character formatting deals with changing the text attributes like its font, font style, colour, size, etc.
- ▶ Paragraph formatting features are collectively applied on all the lines of paragraph.
- ▶ Text alignment, line spacing, paragraph spacing, indentations etc. are some of the paragraph formats.
- ▶ Bulleted (Unordered) and numbered (ordered) are the two types of list that can be created in Word document.
- ▶ Borders and shading features adds to the enhancement of text appearance in the document.
- ▶ Graphical objects added in the document helps to enhance the readability of the document.
- ▶ WordArt is a decorative text effect used for adding special text components.
- ▶ Clip Art gallery is a collection of predesigned images that can be added as an object in the document.
- ▶ We can also insert the pictures or images stored in the computer system.
- ▶ Shapes gallery has many types of predesigned shapes which can be quickly added in the document like lines, geometrical shapes, callouts, banners, stars, etc.



Practice Time

A. Tick (✓) the correct option.

1. Character formatting features can be accessed from _____ group on the Home Tab.
(a) Clipboard ☐ (b) Font ☐ (c) Toolbar ☐
2. _____ option aligns the text evenly between left and right margins.
(a) Left ☐ (b) Right ☐ (c) Justify ☐
3. _____ type of list appears in ascending order.
(a) Bulleted ☐ (b) Numbered ☐ (c) Both ☐



4. _____ gallery is a collection of predesigned images that can be added as an object in a document.

(a) Word Art



(b) Creating list



(c) Clip Art



5. _____ is the major constituent of the document.

(a) Text



(b) Cell



(c) Table



B. Write 'T' for true and 'F' for false statements.

1. Formatting is the general arrangement of the contents of documents.



2. Character attributes deals with the cell attributes.



3. Creating list are of two types.



4. There is only one font size in the word document.



5. Underline option is used to show text in tilted style.



C. Fill in the blanks.

1. _____ is a character formatting features.

2. _____ numbering style is used by default to make an ordered list.

3. _____ is the default spacing between the lines of a paragraph.

4. _____ is a collection of built-in images.

5. _____ is the type of object which is inserted using WordArt.

D. Answer the following questions.

1. Explain the character formatting.

2. Explain how you can change the Font Type.

3. What is the difference between Bulleted and Numbered list?

4. What is line spacing? How much line spacing is added by default in Word document?



5. What is WordArt?

6. Write down the steps of Inserting a picture in word document.

Fun Time



► Given below are the formatting tools available in the Home tab of Microsoft Word 2010. Identify and write their names in the space provided here with.

a.  _____

b.  _____

c.  _____

d.  _____

e.  _____

f.  _____

g.  _____

h.  _____

i.  _____

j.  _____

k.  _____

l.  _____

Lab Time



► Design a poster for the school information board, using Microsoft Word 2010 on the topic 'School fete' as per the details given below.

School Fete		Instructions	
Special Attractions	Eatables	Heading font	— Lucida Handwriting
Dance Competition	1. Dosa	Font size	— 24
Games	2. Chowmein	Color	— Blue
Magic show	3. Chhole Bhature	Text font	— Comic Sans
Quiz	4. Bread pakore	Font size	— 14
31st December		Text color	— Red

Make it attractive with the help of other formatting options like, borders and shading, etc. Take a print out of it and submit to your computer teacher.



Type the following document in Microsoft Word 2010 and save it as 'Food':

Our Food

We need food everyday. We eat food to get energy to grow and to keep ourselves healthy. Different foods have different nutrients. Some foods provide us energy and others help us to grow.

The six nutrients of foods are:

1. Carbohydrates
2. Fats
3. Vitamins
4. Minerals
5. Proteins
6. Water

- Bring the heading 'Our Food' in center and give it bold effect and red colour. Give red border and sky- blue shading to it. Change its Font to 'Arial Black' and Font size to 20.
- Justify the paragraph 'We need food....' and give a line spacing of 2.0 to it.
- Give blue colour and underline effect to the line: 'The six nutrients....'.
- Use Roman numbers to make numbered list of six nutrients.
- Give Italic effect and green colour to the six nutrients.
- Choose the border style as "apples" form Art option.
- Save the file and close it.

Brain Booster



- Illustrations help to make the document more readable. How?

Project Time



- Your teacher has asked you to make a collage of different food items. Select the category food in Clip Art gallery and insert pictures of different food items. Make a collage with these pictures. Mention the names of all the food items using WordArt.

Teacher's Note : Tell the students, the benefits of inserting various type of objects in the document and how these illustrations help to enhance the readability of the document.



Editing and Formatting in MS PowerPoint 2010

5.



This chapter includes



- ▶ Inserting Text
- ▶ Editing the Text
- ▶ Moving the Text
- ▶ Copying the Text
- ▶ Deleting the Text
- ▶ Text Formatting
- ▶ Paragraph Formatting
- ▶ Applying Readymade Theme
- ▶ Replacing a Specific Font

From the previous classes, you know that MS PowerPoint is the most commonly used presentation software. It is widely used in schools to make a lesson more interesting to the students. It is also used in offices to make business presentations. There are slides in a presentation and those can be arranged in different ways to make it highly effective. PowerPoint allows you to give your presentation in different themes to enhance the effects.

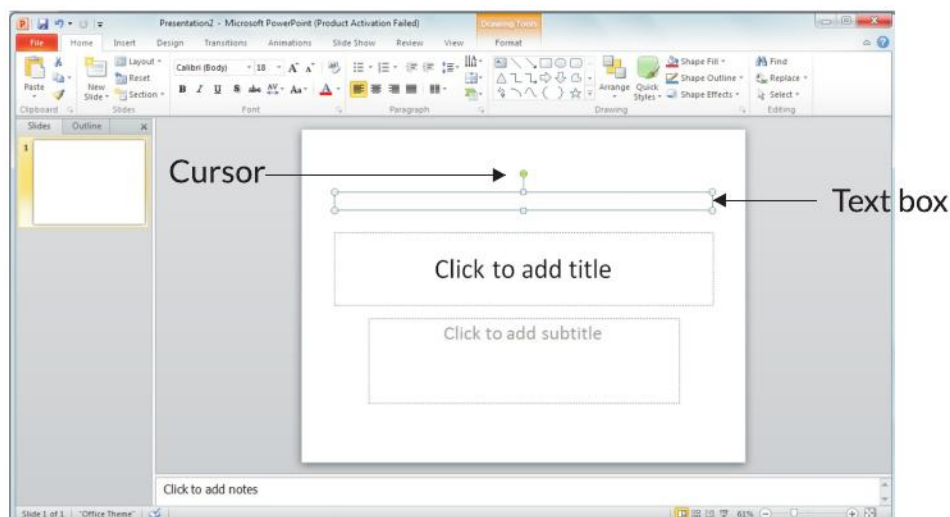
Lets learn about some of the features of MS PowerPoint.

Inserting Text

PowerPoint provides a Text tool which can be used to create additional text placeholders on the slide, if needed.

To create the text box:

- ▶ Click on the **Text box** () tool in the **Text** group on the **Insert** tab.
- ▶ Drag the mouse pointer on the slide where you want to add the text.

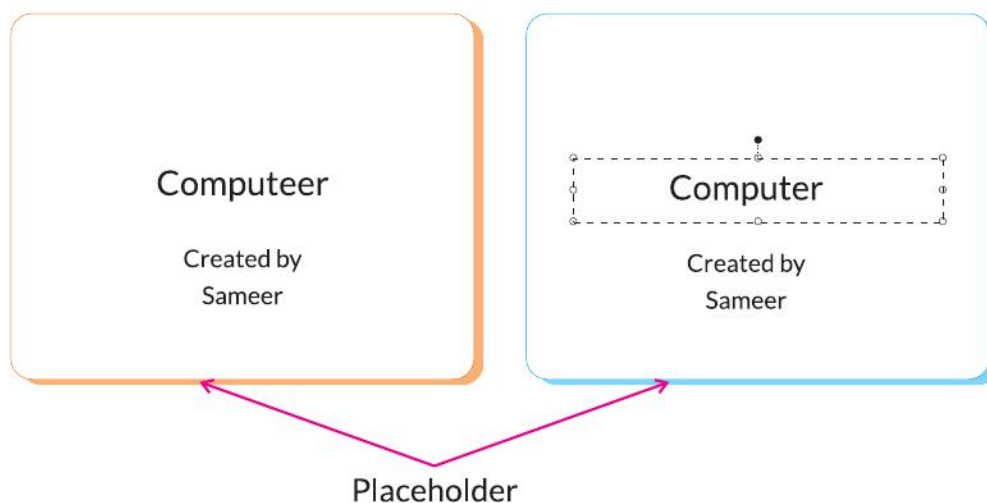


- ▶ Text box will appear with a blinking cursor. Enter the text as needed. The text box expands automatically to the next line if the entered text is more than the size of text box.

Editing the Text

Text entered in a placeholder can easily be modified like this:

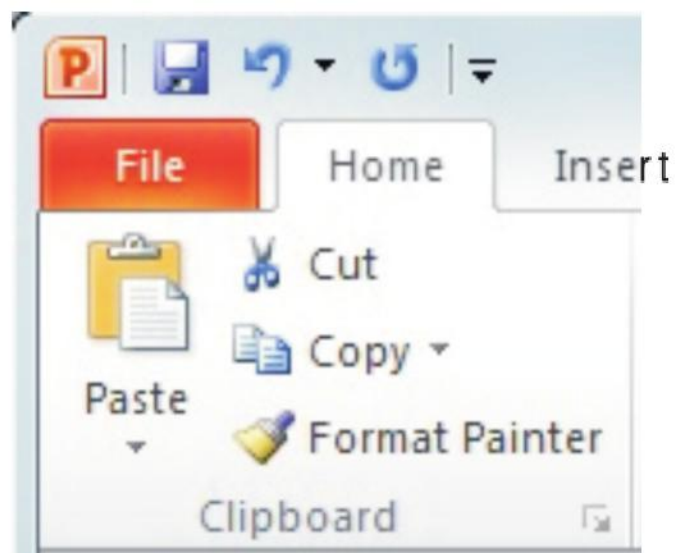
- ▶ Select the slide to be edited.
- ▶ Click on the text object to be modified.
- ▶ Placeholder will appear with the cursor in it.
- ▶ Make change as required and click outside the placeholder.



Moving the Text

This feature can be used when the text entered is to be moved at some other location.

- ▶ Select the text in the placeholder to be moved.
- ▶ Click **Cut** ( Cut) option in the **Clipboard** group on the **Home** tab.
- ▶ Position the pointer at new location and click **Paste** () option in the **Clipboard** group.
- ▶ The selected text will move to the new location.



Copying the Text

You can also copy the text object to the other locations on the slide.

To copy the text:

- ▶ Select the text to be copied and click **Copy** ( Copy) option in the **Clipboard** group.
- ▶ Click to select the location where text is to be copied.
- ▶ Click **Paste** () option to copy the selected text at the new location as well.

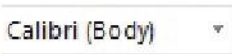
Deleting the Text

To delete the text object on the slide:

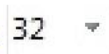
- ▶ Select the text to be deleted. If you want to delete the entire placeholder along with the text, click on its border to select the whole placeholder.
- ▶ Press **Delete** key on the keyboard.
- ▶ The selected text or placeholder will disappear from the slide.

Text Formatting

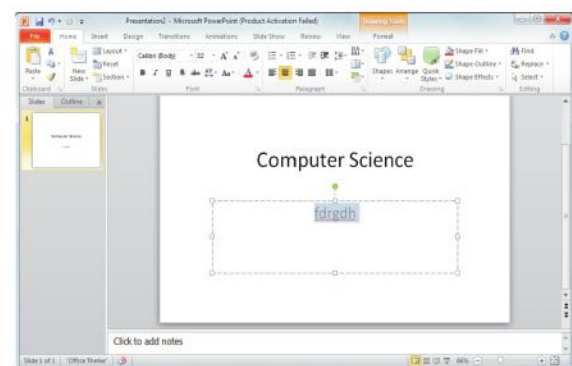
To make the written text more attractive the text formatting feature is used. These options can be accessed from the **Font** group on the **Home** tab. So let us see the use of these options one by one.

Font List 

This option can be used to change the current font type.

Font Size 

This option helps to change the text size. You can increase or decrease the size as required.



Font Group

Bold 

This option can be used to add weight to the select text by making it bolder.

Underline 

This button adds a line under the selected text.

Italic 

It gives slanting effect to the text.

Shadow 

This formatting option adds shadow effect to the selected text.



Font Color

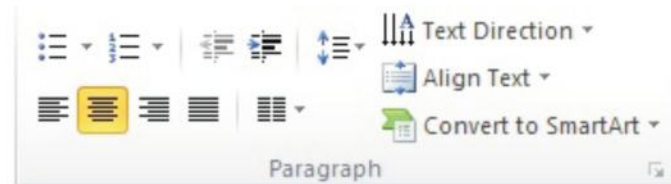
This option is used to change the current text colour with the newly selected colour from the list.

Increase/Decrease Font Size

These two options can be directly used to increase or decrease the font size with respect to the current size by simply clicking on these.

Paragraph Formatting

If we are writing a paragraph, we can apply the paragraph formattings like this:



Paragraph Group

Text Alignment

There are four text alignment options to align the text in the placeholder.

Left () : Aligns the text towards left side of the document.

Right () : Aligns the text towards the right side.

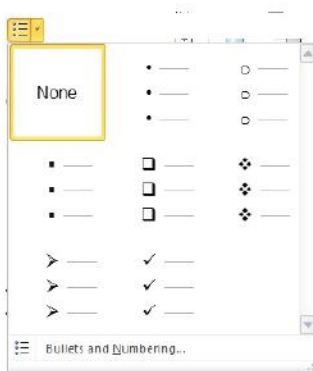
Center () : Aligns the text between left and right edges.

Justify () : Arranges the text evenly between left and right edges of the placeholder.

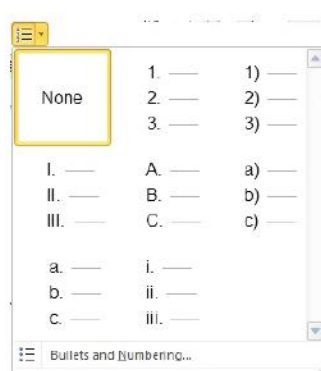
Bullets and Numbering

While writing in MS Word to make points, the Bullets and numbering options are used. These two options on the **Paragraph** group can be used to create ordered (Numbered) or Unordered (Bulleted) lists.

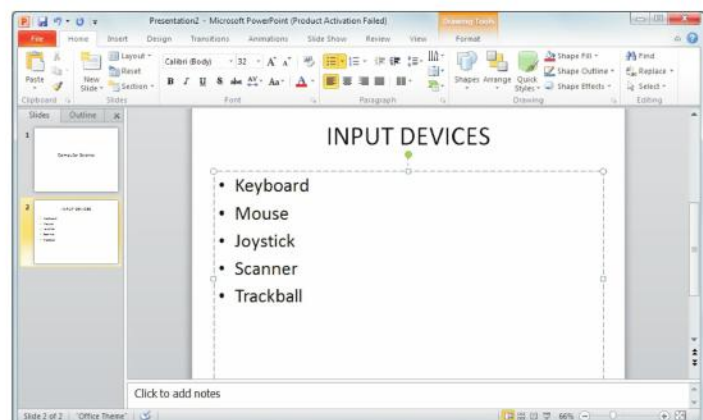
Bullets () option helps to select a symbol which is to be used for marking the list items in an unordered list while **Numbering** () option can be used to select numbering style for creating an ordered list of items.



Bulleted List Options

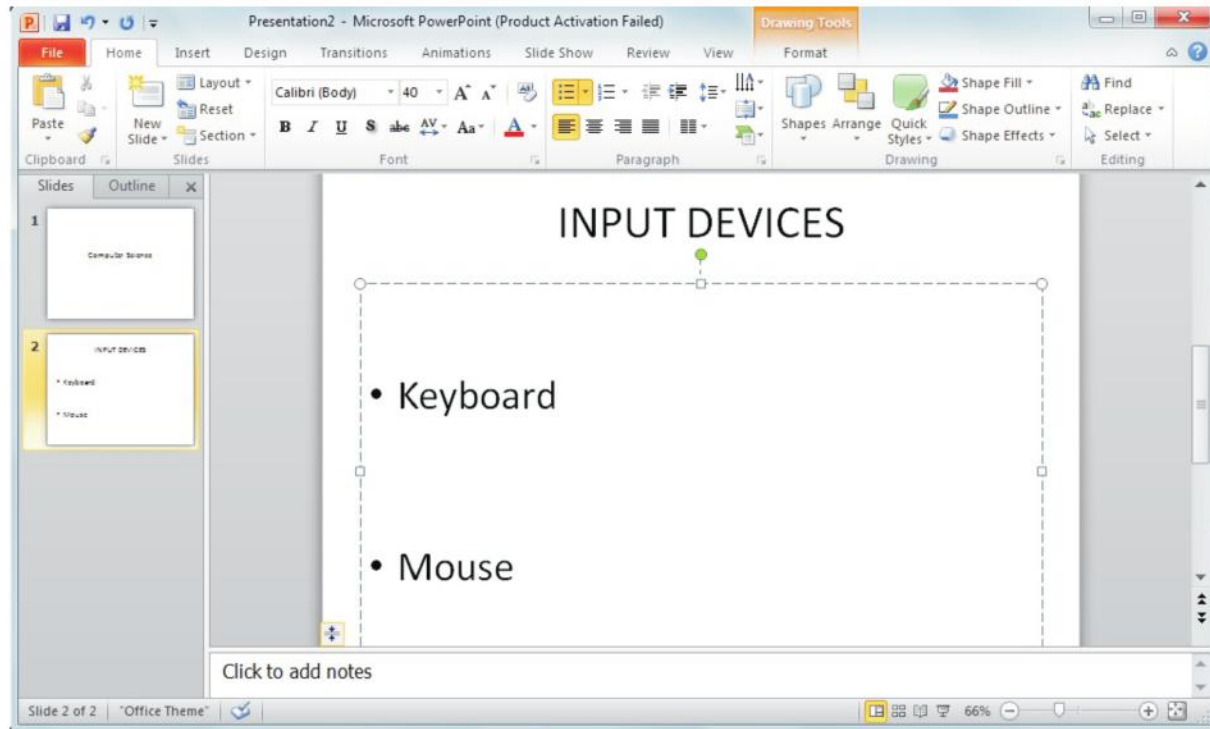
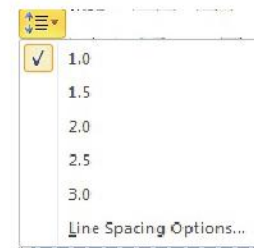


Numbering List Options



Line Spacing

This option is available in paragraph group which is used to change the spacing between the lines of a paragraph. By default, the Line Spacing is 1.0 which can be changed as needed by selecting the specific option from the extended list.



Applying Readymade Theme

You can make a boring black and white slide attractive and colourful. Themes provide the readymade format for the slides which can be used to quickly change the look of your presentation. Each theme has a specific set of fonts, colours and effects along with a background style which can be modified, if needed.

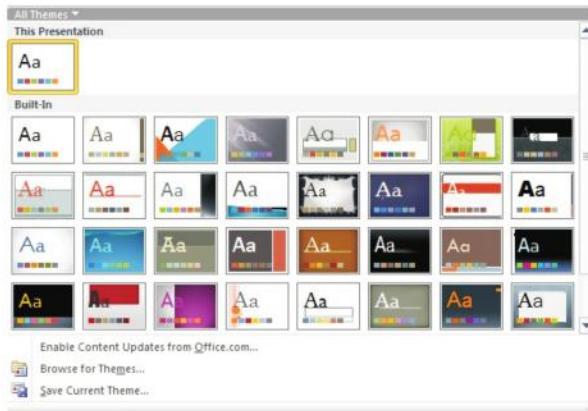
To apply the theme on the slides:

- On the **Design** tab in the **Themes** group select the document theme you want to apply.



- To preview, how the current slide looks like with a particular theme, rest the pointer on its thumbnail in the ribbon.





Computer

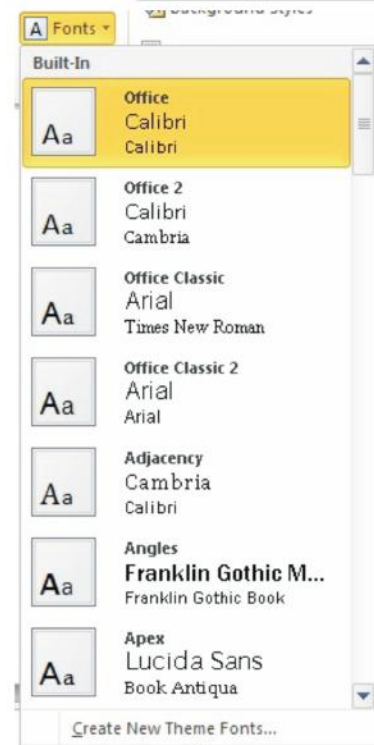
Created by
Sameer



Modifying Theme

The readymade themes can be modified. If you want to make some changes in a predesigned theme before applying it to your presentation, PowerPoint allow you to modify its fonts, colours or line and fill effects like this:

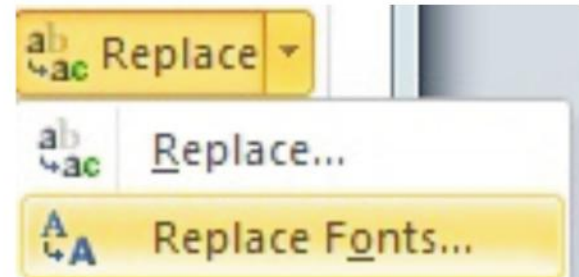
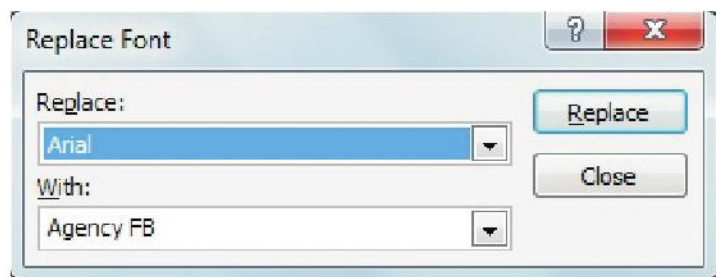
- To change the colour scheme of applied theme, click the **Colours**  drop button on the right side of **Themes** group. Roll over the other colour schemes in the list to preview them on the slide and click the one which is most suitable for the presentation slide.
- Similarly to change the set of font styles used, click the **Font**  drop button and choose a different font set from the expanded list.
- To change the background style, click the **Background Styles** option in the **Background** group.



Replacing a Specific Font

If you want to replace a specific font used in the slide with some other suitable font type.

- Click the **Replace** () drop button in the **Editing** group on the **Home** tab and choose **Replace Fonts** option in it.
- **Replace Font** dialog box will appear as shown in the figure.



- From the **Replace** list which shows the currently used font, choose the one you want to change.
- In the **With** drop list choose the new font type.
- Click **Replace** button to replace the new font.

Let's Summarise



- Editing and formatting features in PowerPoint are used to customize the presentation and to make it more attractive.
- Text Box tool is used to add text anywhere on the slides by creating additional placeholders.
- Text editing features like copying, moving or deleting the text can be used to modify the text contents.
- Features like text formattings, paragraph formattings, etc. help to highlight and organize the text on the slide.
- Readymade themes can be used to quickly format the slide and its contents with single click.
- Replace Font option in PowerPoint can be used to change the specific font in the presentation.





Practice Time

A. Tick (✓) the correct option.

- Text box tool is used to create _____ text on the slide.
(a) Additional ☐ (b) Edited ☐ (c) Formatted ☐
- Cut option in the _____ group on the Home Tab is used to move the text.
(a) File Tab ☐ (b) Clipboard ☐ (c) Toolbar ☐
- Copy and _____ options are used to copy the text.
(a) Cut ☐ (b) Paste ☐ (c) Both (a) and (b) ☐
- _____ option can be used to change the current font type.
(a) Font list ☐ (b) Font size ☐ (c) None of these ☐
- _____ aligns the text towards left side of the document.
(a) Left ☐ (b) Right ☐ (c) Center ☐

B. Write 'T' for true and 'F' for false statements.

- Editing and formatting features are used to customize the presentation. ☐
- There are no line spacing options available in MS PowerPoint. ☐
- The readymade themes can make the slides attractive. ☐
- No modifications can be made to the predesigned themes. ☐
- We can change the bullet and numbering style anytime even after completing the list. ☐

C. Fill in the blanks.

- Text tool box can be accessed from _____.
- _____ is used to apply readymade themes to slides.
- _____ is the default line spacing of text.
- _____ option is used to change the specific font in the presentation.
- Clipboard group is present in _____.

D. Answer the following questions.

- Write down the steps for editing the text.
- Name the different options of Text Formatting.
- Explain the text alignment options to align the text in the placeholder.
- Which special effect options in the Font group can be used to make the text attractive in the slides?
- What is the use of Replace Font option?



Fun Time



- Look at the following slide layouts offered by Microsoft PowerPoint to create presentation slides. Identify and write the names of these layouts.

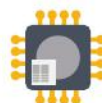


Lab Time



- Collect the information about the India International trade fair organized this year in the Pragati Maidan, Delhi and design a presentation containing slides with following titles:
1. Title slide showing the presentation title and your name.
 2. Objective and Focus of the trade fair.
 3. Theme of the trade fair
 4. Opening ceremony.
 5. Participating states and countries.
 6. Expected business.

Brain Booster



- Why do we need to apply the formatting features on the presentation contents? Does it help only to make it attractive or something else? Explain.

Project Time



- Discuss the benefits of using readymade themes for the formatting of PowerPoint slides over the customized formatting option in a group of four students each and then discuss them collectively in the class.

Teacher's Note : Text being one of the major part of presentation slides help the audience to quickly understand the presentation contents if it is properly formatted. So tell the students about the benefits of formatting features and what kind of formattings are needed in various situations.



Introduction to Microsoft Excel 2010



This chapter includes



- ▶ Starting Microsoft Excel 2010
- ▶ Components of Microsoft Excel File
- ▶ Editing Cell Value
- ▶ Using Name Box to Locate a Cell
- ▶ Closing a Workbook
- ▶ Types of Data
- ▶ Entering Data in Excel Worksheet
- ▶ Moving around in the Worksheet
- ▶ Saving a Workbook
- ▶ To Close Microsoft Excel Application

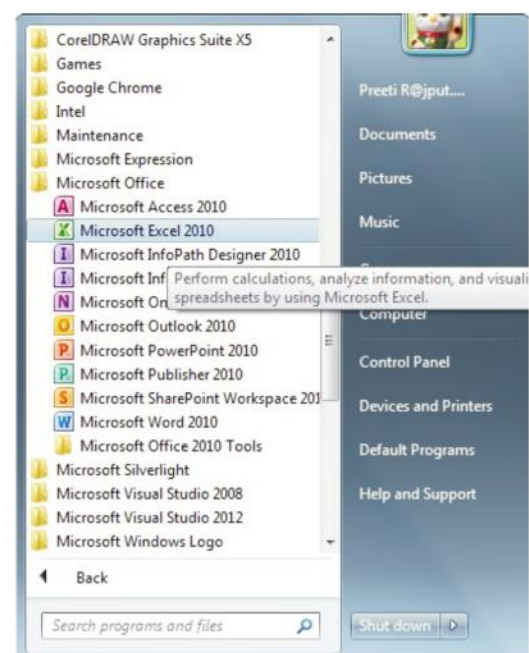
Microsoft Excel 2010 is an important member of MS Office 2010 and one of the most popular Windows based spreadsheet application which is used for storing data in a tabular form, performing simple as well as complex calculations and for creating graphs, etc. MS Excel is a spreadsheet software which helps us to calculate and analyse data. It contains data in row and column form.

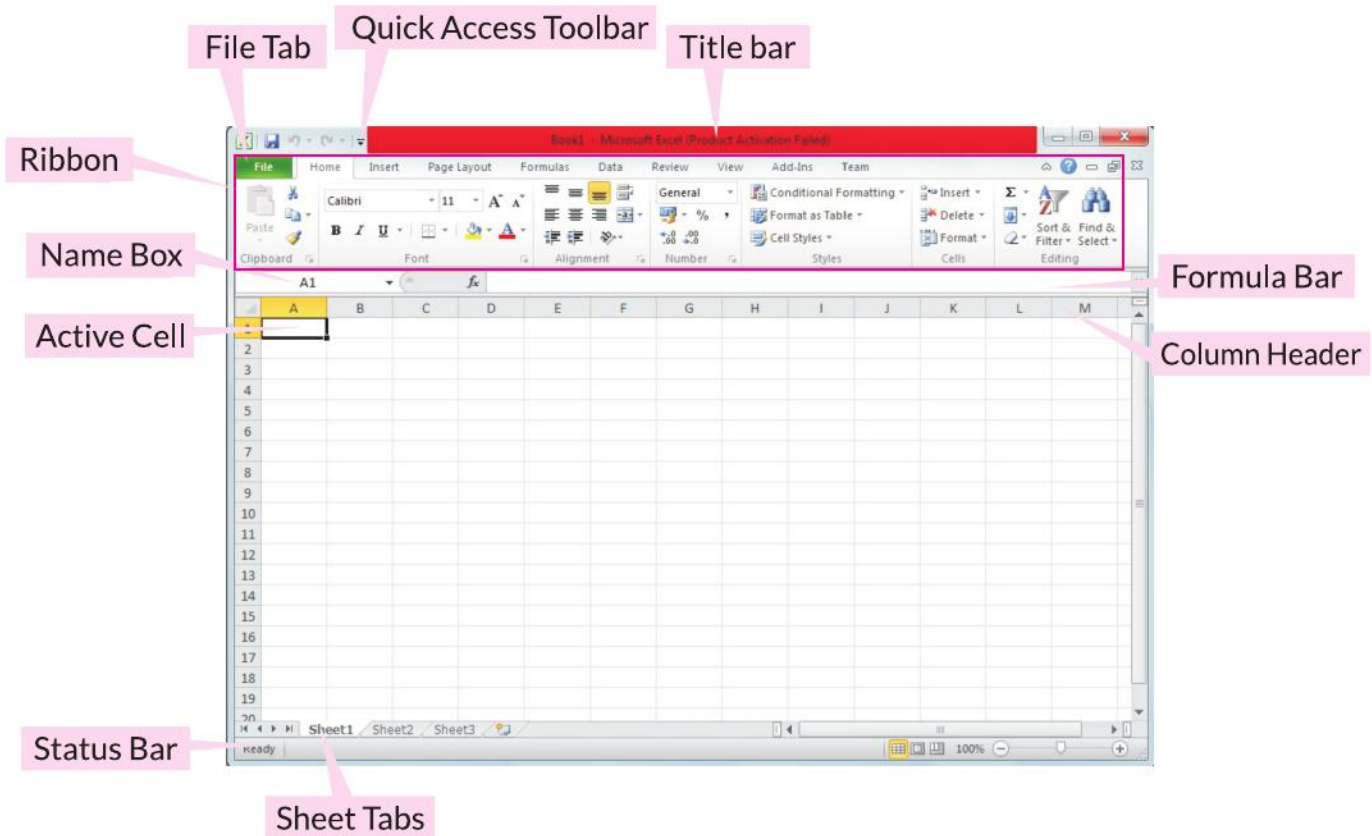
There are many other spreadsheet applications available like OpenOffice.org Calc, Lotus 123, Quattro Pro, VisiCalc, etc. but among all these Microsoft Excel is the most powerful spreadsheet application for creating electronic worksheet. So let us start with this application.

Starting Microsoft Excel 2010

To start Microsoft Excel:

1. Click the **Start** button.
2. Choose **All Programs**.
3. Click on **Microsoft Office**.
4. Choose **Microsoft Excel 2010**.





Microsoft Excel 2010 window will appear on the screen as shown in figure with a blank workbook having three worksheets by default. The default workbook name as Book1 which can be changed while saving the workbook.

Components of Microsoft Excel 2010 Interface

Various components of Microsoft Excel 2010 interface which are used to work with electronic spreadsheet are as follows:

File tab: The new design of Microsoft Excel includes the File tab replacing the Microsoft Office button included in MS Office 2007. It gives access to backstage view where we can manage our Excel data files.

Title bar: It appears at the top of Excel window which shows file name and application name.

Quick Access Toolbar: Quick Access Toolbar is used to access frequently used features in Excel like save, undo, redo, etc. It is present in the title bar.

Ribbon: It gives access to the Excel commands available in the form of buttons and icons. Located at the top of your worksheet having seven tabs in it which are further divided in groups containing options to create and work with spreadsheet.

Formula Bar: Mainly used to create formulas for performing arithmetic or logical operations. Data can be entered directly in a cell using formula bar.



Name Box: Displays the cell address of the active cell.

Sheet Tabs: The Sheet tabs are located at the bottom of the window. They display the names of the sheets being used in the current workbook. The active sheet name is displayed as an open box and from here only we can open other worksheets when required.

Status Bar: The Status Bar at the bottom of the window displays information about the command being used and the status of the worksheet being processed.

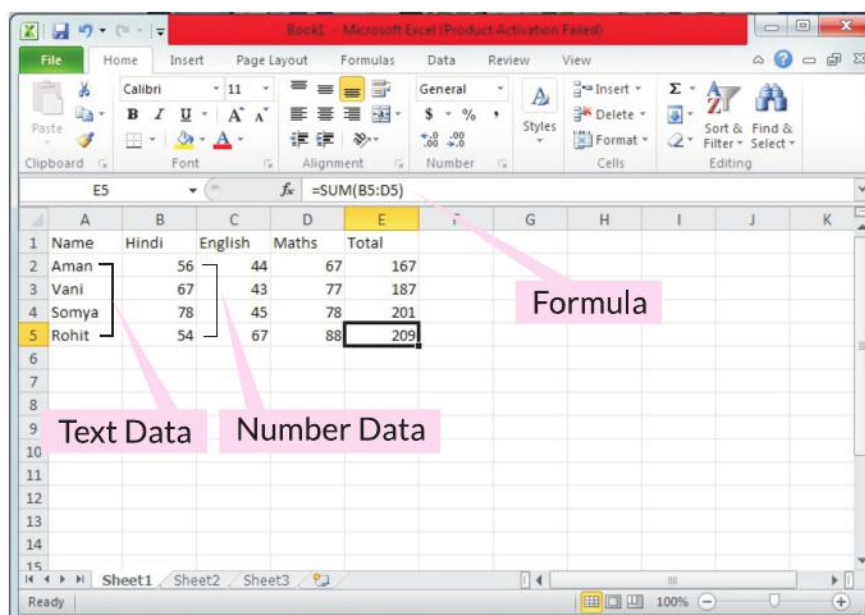
Types of Data

There are three different types of data that can be entered in Microsoft Excel worksheet. These are:

Text: Text data is combination of alphabets and numbers which cannot be used for mathematical calculations. By default text data is left aligned in a cell.

Number: Number data is formed using the digits 0–9 on which the arithmetic calculations can be performed. By default numbers are right aligned in a cell.

Formula: You can create formulas in Excel spreadsheet for easy and quick calculations using cell values or cell reference. The formula results are right aligned by default within the cell boundary.



Components of Microsoft Excel File

The following are the various components of Microsoft Excel File:

Workbook

An Excel document is called a **Workbook** which is the basic Excel file. It is a collection of multiple worksheets (also called spreadsheets). There are 3 worksheets by default in a workbook. The Excel workbook is saved with .xlsx extension.

Quick Question



What is default name of a workbook in Microsoft Excel 2010?

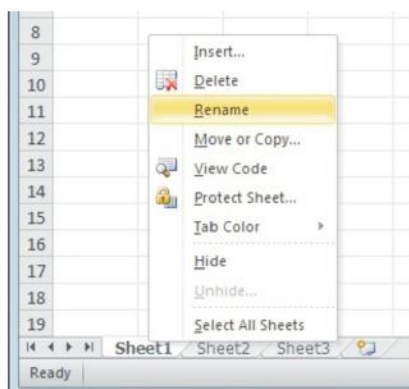
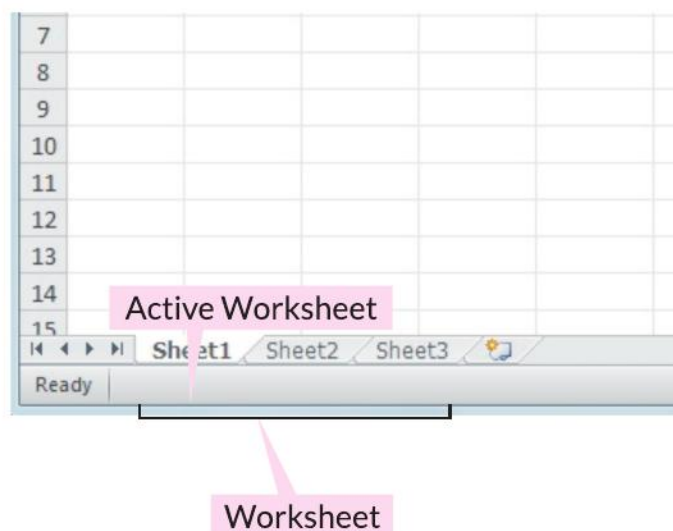


Worksheet

Worksheet is an individual page of Excel Workbook also known as **Spreadsheet**. It is made up of rows and columns, intersecting each other to form cells.

The default name of the worksheets are Sheet1, Sheet2 and Sheet3 which can be renamed suitably as needed.

These sheet names are displayed in the sheet tab area. You can switch between these worksheets by clicking on their respective tabs in this area.



To give a meaningful name to the worksheet:

- Double click the sheet name in the sheet tab.

Or

Right click on the sheet name in the **Sheet** tab and choose the **Rename** option in the shortcut menu.

- Enter new name and press **Enter** key.

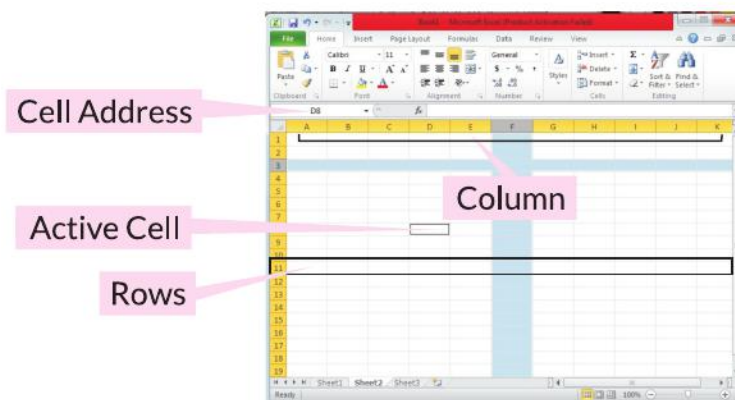
Rows

Worksheet is a combination of intersecting rows and columns. Rows are represented horizontally from top to bottom in the worksheet. There are 1,048,576 rows in a worksheet which are numbered from 1 to 1,048,576.

Columns

Columns are represented vertically from left to right.

Columns are labelled as letters A, B, C and so on upto XFD. There are total 16,384 columns in a worksheet.



Cells

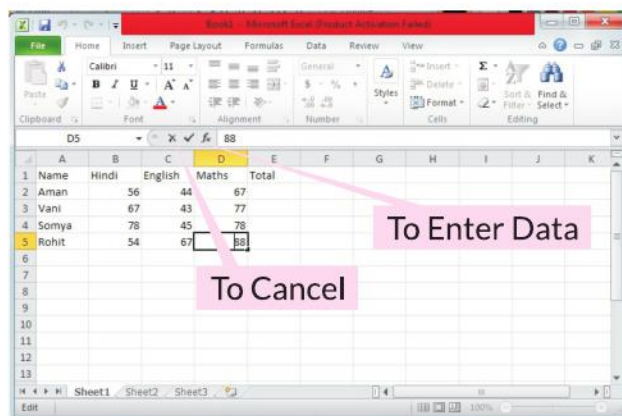
Cell is the smallest unit of the worksheet formed by the intersection of a row and a column. Each row in a worksheet intersects each column to form rectangular blocks called **Cells**. The data in a worksheet is organized in these cells. A cell with a highlighted cell boundary is called **Active cell**. Data can be entered or edited in the active cell only. To make any cell active, you can click that cell or by using arrow keys you can move the highlighted boundary to any of the worksheet cell.

Cell Address

Each cell on the worksheet has a unique address which helps to refer it in the worksheet. The address of a cell is column name followed by the row number which intersects to form that cell. For example, if row no. 1 intersects column A, then the cell formed with the intersection of these two will be referred as A1. Using these cell addresses you can easily locate a value in the worksheet or they can be used for creating referential formulas.

Entering Data in Excel Worksheet

Data in the Excel worksheet can be entered simply by selecting the cell and then typing the required values in it. When the data is typed, it appears in the active cell as well as in the **Formula Bar**. You will also notice two small buttons in the Formula bar (✕) and (✓). If you click (✕) button or Esc. Key while typing, the entered data will be erased from the cell. Pressing (✓) button or **Enter** key will store the data in the active cell. You can enter upto 32767 characters of data in a cell.



Quick Question

Name the key used to cancel data entry.

Editing Cell Value

The entered data in a cell can be modified in two ways:

- By replacing the previous cell value with the new value.
- By partially modifying the cell values.

To replace the contents of a cell with the new value:

- Select the cell containing data to be replaced.
- Enter the new cell value.
- Press **Enter** key or click () button.

To partially modify the cell contents:

- Select the cell whose contents have to be modified.
- Click in the **Formula bar**
Or
- Double - click the cell.
Or
- Press **F2** function key.
- The cursor will appear in the cell.
- Now, change the cell contents as needed.
- Press **Enter** key or click () button in the **Formula bar**, to store modified value in the cell.

Moving around in the Worksheet

The MS Excel worksheet is very large. So, it will be difficult to find the inserted data from the 16,384 columns and 1,048,756 rows, Each row intersect each column in the worksheet forming 17,17,98,69,184 cells to store worksheet data.

To make it easy to move around quickly in such a large worksheet, the following keys are used:

Key	Function
Left Arrow	Move one column left.
Right Arrow or Tab key	Move one column right.
Up Arrow	Move up by one row.
Down Arrow or Enter	Move down by one row.
Page Up	Move up by one screen.
Page Down	Move down by one screen.
Home	Move to the first cell of the row.

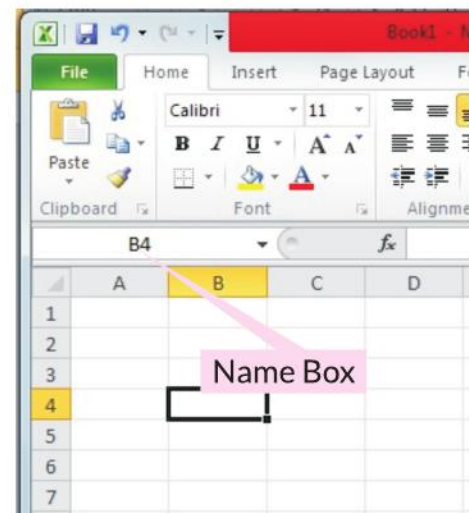


Key	Function
Ctrl + Down Arrow	Move to the last row, i.e., 1,048,576 in the worksheet.
Ctrl + Up Arrow	Move to the first row, i.e., 1.
Ctrl + Left Arrow	Move to the first column, i.e., A.
Ctrl + Right Arrow	Move to the last column, i.e., XFD.
Ctrl + Home	Move to the cell A1.
Ctrl + End	Move to the last used cell in worksheet.

Using Name Box to Locate a Cell

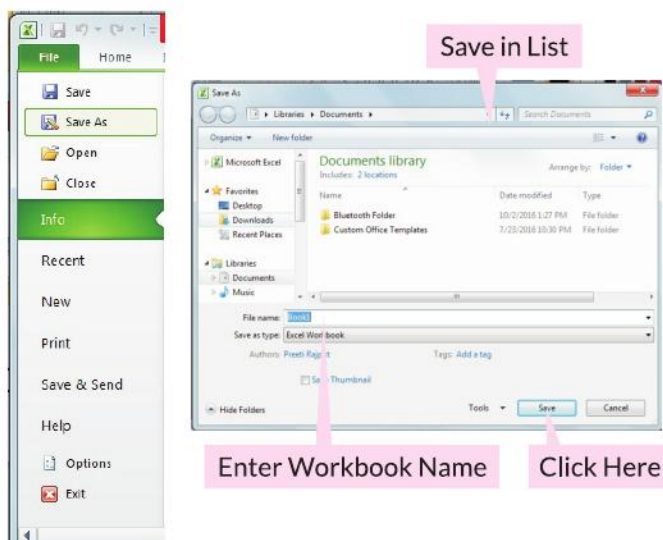
If you want to go to a specific cell in the worksheet, you can make use of Name Box, located at the left side of the Formula bar.

- Click in the **Name Box** to select the address of current cell in it.
- Now enter the cell address where you want to move, e.g., E21.
- Press **Enter** key to reach the destination cell.



Saving a Workbook

You can enter the data in different worksheets of a workbook. All these worksheets are saved together as a single workbook.



To save the workbook:

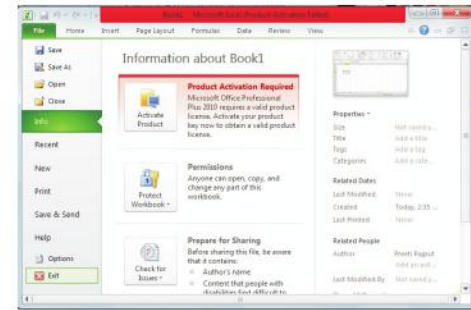
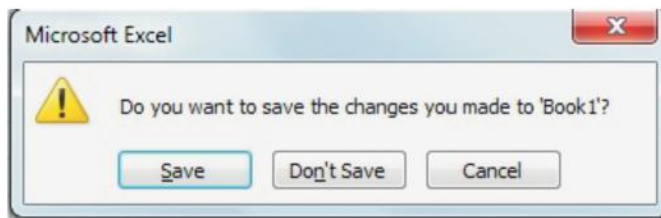
- Click **File** tab and choose **Save As** option in it. **Save As** dialog box appears as shown in figure.
- In the **Save In** list, select the location where you want to save the file.
- In the File Name box, enter the name of your file.
- Click the **Save** button to save your workbook permanently.



Closing a Workbook

To close the workbook after finishing your work:

- ▶ Click **File** tab and select **Close** option in it.
- ▶ If you have not saved your currently opened file, Excel prompts you to save your workbook. Click **Save** to save and close your file. Click **Don't Save** to close the file without saving. Click **Cancel** to terminate the save process.



Fact Bytes



.xlsx extension is automatically added to the file name when it is saved.

To Close Microsoft Excel Application

- ▶ Choose **Exit** option in the **File** tab.
- Or
- ▶ Click () button on the right side of the titlebar.



Let's Summarise



- ▶ Microsoft Excel 2010 is a spreadsheet program used for storing data, performing calculations and creating graph, etc.
- ▶ Worksheet is a collection of cells formed with the intersection of horizontal rows and vertical columns.
- ▶ Workbook is an Excel document. It is a collection of worksheets.
- ▶ There are three worksheets added by default in an Excel workbook.
- ▶ In an Excel worksheet, there are 1,048,576 rows and 16,384 columns.
- ▶ Rows are numbered as 1, 2, 3,..... 1,048,576 whereas columns are labelled as A, B, C..AA, AB... XFD.
- ▶ Three types of data can be entered in Excel worksheet i.e., Text, Number and Formula.
- ▶ Cell address is column name followed by row number.
- ▶ Name box in the formula bar can be used to locate any cell on the Worksheet.





Practice Time

A. Tick (✓) the correct option.

- _____ is located at the bottom of MS Excel window.
a. Name Box ☐ b. Formula Bar ☐ c. Status Bar ☐
- _____ appears at the top of MS Excel window.
a. Title Bar ☐ b. Ribbon ☐ c. Formula Bar ☐
- _____ is the smallest unit of worksheet.
a. Cell ☐ b. Rows ☐ c. Columns ☐
- When the data is typed, it is also appeared in _____.
a. Cell ☐ b. Formula Bar ☐ c. Columns ☐
- Exit option is present in _____.
a. Formula Tab ☐ b. File Tab ☐ c. View Tab ☐

B. Write 'T' for true and 'F' for false statements.

- Quick Access Toolbar is present in the Title Bar. ☐
- The sheet tabs display the names of the sheets being used in the current workbook. ☐
- The Save As option is present in File Tab. ☐
- There are 17 cells in MS Excel worksheet. ☐
- Ctrl + Home is used to move to the cell A1. ☐

C. Fill in the blanks.

- _____ contains data in the form of rows and columns.
- There are _____ rows in a worksheet.
- Press _____ key to move one column right.
- An Excel document is called _____.
- There are _____ types of data that can be entered in Excel workbook.

D. Match the following.

- | | |
|----------------|---|
| 1. Page Up | a. To move in the last used Cell in Worksheet |
| 2. Page Down | b. To move in the first column |
| 3. Home | c. To move in the cell A1 |
| 4. Ctrl + Home | d. To move up by one screen |
| 5. Ctrl + End | e. To move down by one screen |



E. Answer the following questions.

1. What is Microsoft Excel? What is its use?

2. Write any four shortcut commands that can be used in MS Excel.

3. Describe the components of MS Excel.

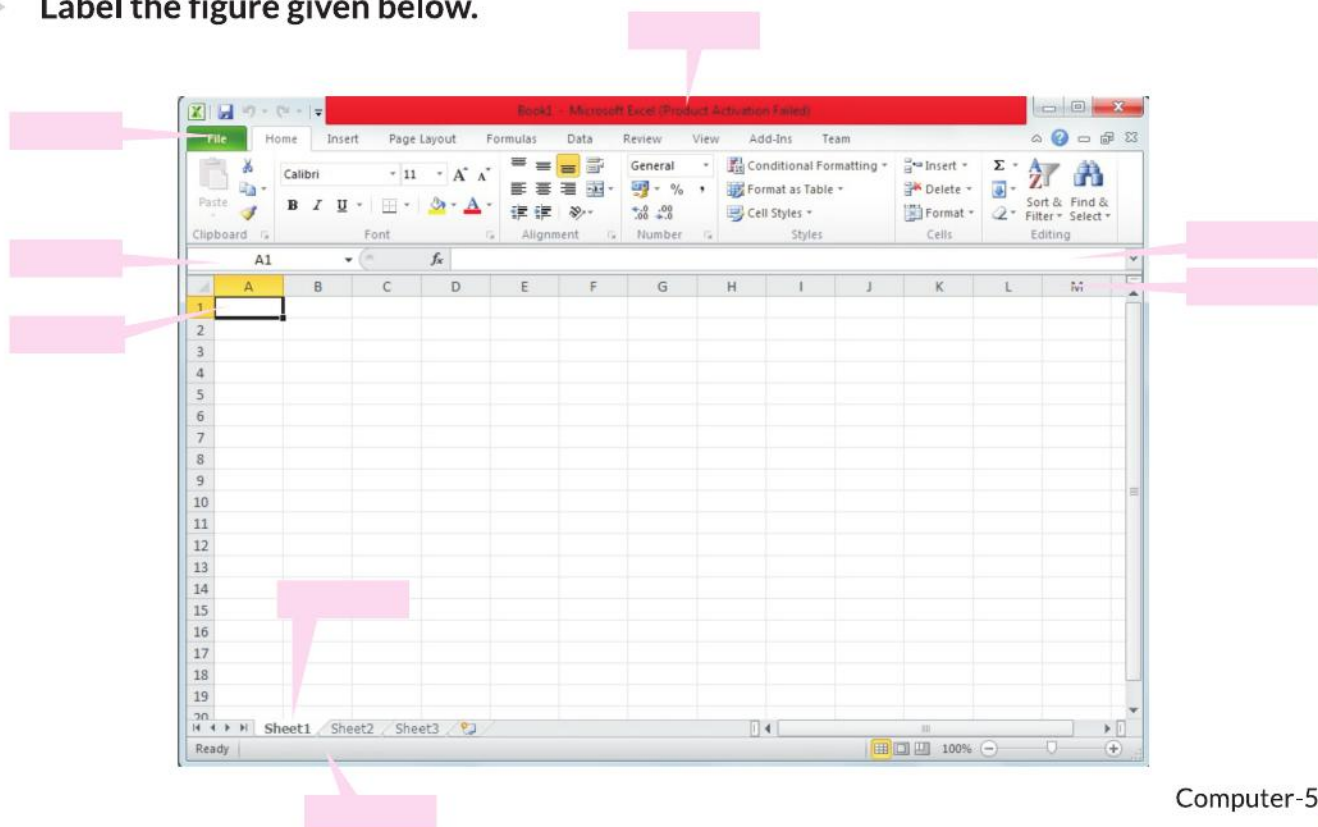
4. Write down the steps to Rename a Excel workbook.

5. Write the steps for saving a Excel workbook.

Fun Time



► Label the figure given below.



Lab Time



- **Start Microsoft Excel** Click on the Start button → All Programs → Microsoft Office → Microsoft Excel 2010.

A blank workbook Book1 will open in Excel window.

Type the following data in it:

S.No	Item	Price	Qty.	Total Amount
1	Pencil	3	10	30
2	Eraser	5	20	100
3	Colours box	20	10	200
4.	Pen	10	5	50

- Save the workbook (Click File tab → Save As) as 'Bill'.
- Change the name of worksheet 'Sheet1' to 'Purchase' by double-clicking the sheet name in the Sheet tab.
- Replace the item name 'Pen' with 'CDs' by using overwrite method.
- Change the CDs price to 15 and total amount to 75 using partial modification method.
- Save the workbook again and Close Microsoft Excel.

Brain Booster



- How is it beneficial to use electronic spreadsheet over the manual worksheet?

Project Time



- Visit any computer shop and ask the price of different parts of computer. Make a table in Microsoft Excel using given format. Fill in the price column to add up the total cost for a computer. Compare your price with your

Parts of a Computer		
S.No.	Parts	Price
1.	Keyboard	_____
2.	Mouse	_____
3.	Monitor	_____
4.	Speakers	_____
5.	CPU	_____
	TOTAL	_____

Teacher's Note : Tell the students about the various advantages of spreadsheet by giving different examples. Also explain them the benefits of using electronic spreadsheet programs over manual spreadsheets. Demonstrate them how to move around in a spreadsheet.



Algorithm of Flowchart

7.



This chapter includes



➤ Algorithm

➤ Flowchart

Computer is a machine. To make it work, it has to be given instructions called programs. We can say that a program is a set of instructions given to a computer to solve a particular problem. The first step in generating a computer program is to understand the problem and then put down the steps in sequence to get the result.

Writing a program involves three steps.

- Writing an algorithm
- Designing a flowchart based on an algorithm
- Converting the flowchart into a program

Fact Bytes



Algorithms and flowcharts are tools which make development of programs easier.

Algorithm

An **algorithm** is a representation of general instructions given to solve a particular problem. It is written in a simple language and proper sequence. It is important to understand the nature of a problem before solving it.

Rules for Writing Algorithms

There are some guidelines for writing an algorithm.

- Give statement numbers as Step 1, Step 2, etc.
- Always begin with Start.
- Write each instruction in a separate line.
- Write the function of each statement in a separate line.
- Write Stop at the end of the algorithm.

Example

- If you want to write a program to calculate the speed of a car, the step would be written as:
Step 1: Enter the time taken and distance.
Step 2: Divide the distance by time.
Step 3: Find the speed.



So, algorithm is a rough estimate to identify the input, process and output. In the algorithm, Step 1 is identified as input, Step 2 as process and Step 3 as output.

Let's see another example of algorithm and flowchart that writes a program to input two numbers and print the sum.

Algorithm

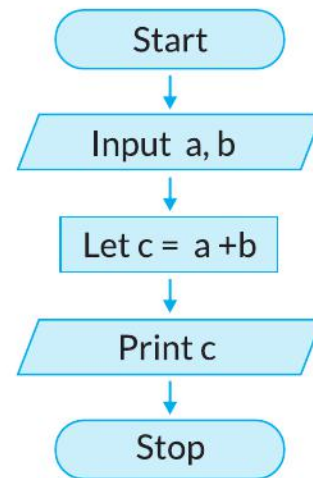
Step 1: Start

Step 2: Input a, Input b

Step 3: Let $c = a + b$

Step 4: Print c

Step 5: Stop



Algorithms are often broken down into smaller chunks called sub-procedures. This is both so that they are easier to read, and also because then parts of the algorithm can be reused.

Example

- Write an algorithm to multiply two numbers.

Step 1: Start

Step 2: Input first number

Step 3: Input second number

Step 4: Multiply the two numbers

Step 5: Print the result

Step 6: Stop

Example

- Write an algorithm to find the area of rectangle.

Step 1: Start

Step 2: Input the length of rectangle (L)

Step 3: Input the breadth of rectangle (B)

Step 4: Calculate the area of rectangle ($L \times B$)

Step 5: Print the area (A).

Step 6: Stop.

Example

- Write an algorithm to find the perimeter of square.

Step 1: Start

Step 2: Input the side of the square (S)

Step 3: Calculate the perimeter of square ($P = 4 \times S$)

Step 4: Print the perimeter (P)

Step 5: Stop.



Advantages of Algorithm

There are many advantages of writing algorithms for solving any problem.

Some of them are as follows:

- They are easy to understand.
- They are easy to implement.
- They describe the steps that should be taken to solve a problem.
- They help eliminate any errors in our problem solving logically.
- They are independent of any computer language.
- They can be easily converted into flowcharts and then into computer programs.

Flowchart

A **flowchart** is a diagrammatic representation of a program. Building a flowchart is like putting a puzzle together. There are standard pieces. The standard action shapes include terminal, process, input/output, decision and junction.

This graphical representation plays a vital role in the programming of a problem and is helpful in understanding the logic of complicated and lengthy problems.

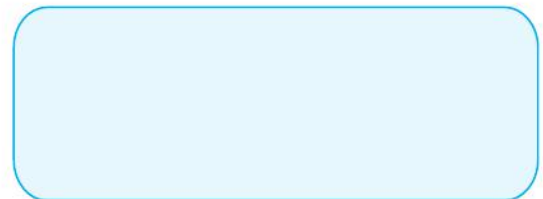
Once the flowchart is drawn, it becomes easy to write the program in any high level language. Flowcharts are usually drawn using some standard symbols.

Rules for drawing or/and formatting a Flowchart

- A flowchart always begins with the Start symbol and ends with the Stop symbol.
- The connecting arrow between two steps helps to indicate the logical sequence from one step to the other.
- Every symbol of a flowchart represents a particular action.
- Arrowheads indicate the flow of data.
- A connector is used to connect breaks in the flowchart. It also joins one page to another, bottom of the page to the top of the same page and an upward flow of more than three symbols.

Terminal Box

The **terminal box** is an oval-shaped or rounded rectangle box which shows the beginning and the end of the flowchart. Each flowchart can have only one beginning point and only one ending point. In the beginning of the flowchart START is written and at the end, STOP is written in the terminal box.



Input-Output Box

The **input-output box** is used to show input of the data and also output for the result. This box is parallelogram in shape. For showing input data, INPUT or GET is written and for showing the result, PRINT or DISPLAY is written.



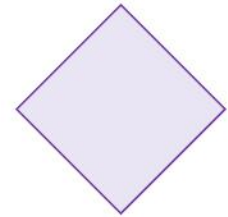
Processing Box

The **processing box** is used to show the processing work. It shows calculations. This box is rectangle in shape.



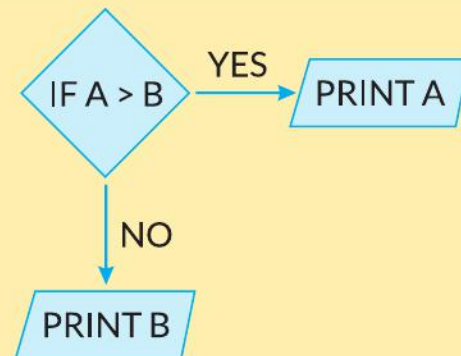
Decision Box

The **decision box** is diamond-shaped and is used to check some condition or compare two values. Each condition can have two answers, YES or NO. This box has two exit points and one entry point.



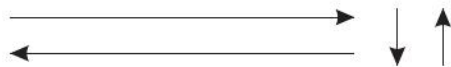
Example

- To check whether $A > B$ and if yes PRINT A else PRINT B, the flowchart will be:



Flow lines (Arrows)

All the flow charting symbols are connected to each other by using the **flow lines**.



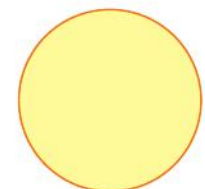
Fact Bytes



Only one flow line can exit or enter the START and STOP symbols, respectively.

Connectors

The **connector** in a flowchart is used for continuing the flowchart to the next page. It is also used to jump the normal flow of a flowchart. The connector is given a name which is written in the centre.



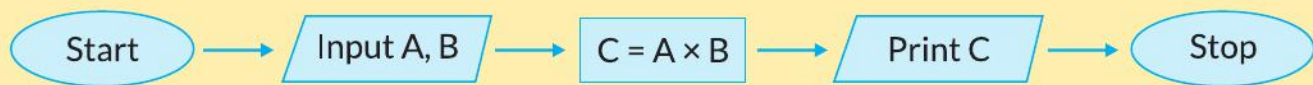
Advantages of a Flowchart

A flowchart provides the following benefits:

- It is usually much easier and faster to draw a flowchart of a problem than to write a program directly.
- Flowcharts are an important aid in the development of the algorithm itself.
- It is easier to understand a flowchart than a program itself.
- A flowchart is independent of any particular programming language. We can develop a program in any language from a flowchart.

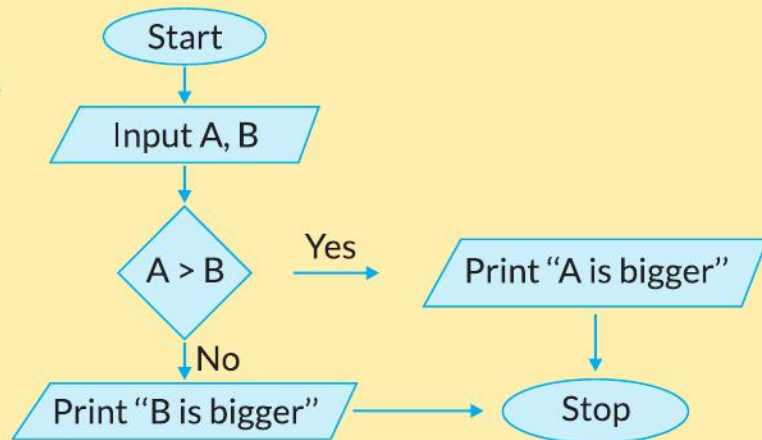
Example

- Draw a flowchart to multiply two numbers A and B.



Example

- To find the bigger of two numbers and display a message also.



Let's Summarise



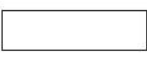
- An algorithm is a sequence of steps written in a simple language to solve a problem.
- An algorithm is a representation of a solution to a problem.
- Flowcharts are very important in computer programming.
- The flowchart is a pictorial representation of the sequence of steps involved in solving a particular problem.
- Flowcharts have some special boxes to show the operations.
- Every flowchart should have a START box as the first box and STOP box as the last box.
- Every flowchart must have only one entry point and one exit point.





Practice Time

A. Tick (✓) the correct option.

1. What is a graphical representation of a program called?
(a) Flowchart ☐ (b) Algorithm ☐ (c) Instruction ☐
2. What is the shape of Input/output box?
(a) Circle ☐ (b) Rectangle ☐ (c) Parallelogram ☐
3. The arrows in a flowchart are called _____.
(a) Flow lines ☐ (b) Connectors ☐ (c) Line segments ☐
4. Which shape is used for the decision box?
(a)  ☐ (b)  ☐ (c)  ☐

B. Write 'T' for true and 'F' for false statements.

1. An algorithm is written in simple language and proper sequence. ☐
2. Flowcharts are not important in computer programming. ☐
3. The algorithms are easy to understand and implement. ☐
4. A flowchart always begins with the START symbol. ☐
5. Process box shows input or output of data. ☐

C. Fill in the blanks.

1. The circle-shaped symbol in flowchart is called _____.
2. An algorithm is written in a _____ language.
3. _____ is the shape of processing box in flowchart.
4. _____ indicate the direction of the instructions in a flowchart.

D. Answer the following questions.

1. Write three steps involved in writing a program.

2. Write any two rules for writing an algorithm.

3. Write any two advantages of an algorithm.



4. What is a flowchart. Write its two advantages?

5. Differentiate between a flowchart and an algorithm?

Fun Time



► Draw the required box in front of each statements. (Here, two options are solved.)

Assign the value 5 to b ($b = 5$), e.g., $b = 5$

- | | |
|--|--------------------|
| a. Calculate the total of two numbers, e.g., Sum = a + b | |
| b. Start the flowchart | c. Read two values |
| d. Print the total | e. Print your name |
| f. Accept a name | |

Lab Time



► Open Word 2010 and draw the flowchart to:

1. Find the area and perimeter of rectangle.
2. Find the area of square.

Brain Booster



► Can the process /action of going to watch movie ,be shown through an algorithm? If yes, how?

Project Time



► Prepare a presentation on — 'Algorithm and Flowchart'.

Teacher's Note : Explain the usage of Algorithm and Flowchart to the students.



Exploring Internet

8.



This chapter includes



- ▶ Internet
- ▶ Birth of the Internet
- ▶ How are so many computers connected?
- ▶ Internet Terms
- ▶ Requirements to Access the Internet
- ▶ Communicating over the Internet
- ▶ E-mail
- ▶ Getting to Gmail

A computer network is simply a connection of two or more computer to exchange information. A small network can be as simple as two computers linked together by a single cable.

A wireless network connects computers without cables. Computer use radio communications to transfer data among each other.

Internet

The Internet is the biggest network of computers in the world that connects millions of computers and computer networks together for the purpose of sharing information and facilitating communication.

The Internet is also called global network of computers. Starting with a few users, it now has millions of users all over the world.

Birth of the Internet

The foundation of the Internet was laid in the 1960s, in the United States of America. In 1969, the first network **ARPAnet (Advanced Research Projects Agency Network)** was created by the US Department of Defense, to connect the computers of the US defense and US Universities like UCLA and Stanford.

- ▶ In 1972, **e-mail** was introduced by **Ray Tomlinson**, a scientist at Cambridge, Massachusetts.
- ▶ In 1973, the ARPAnet went global when the University College of London joined in.
- ▶ The ARPAnet was further joined by the NSFnet and linked to other private networks to allow commercial and governmental activities. This growing network ultimately formed the Internet.



- ▶ In 1983, the word **Internet** was used for the first time.
- ▶ In 1991, Tim Berners-Lee and others, at the European Laboratory for Particle Physics, more popularly known as CERN, developed the **World Wide Web**. This made the Internet much more user-friendly and commercially viable.



Fact Bytes



The early Internet was used by computer experts, engineers, scientists and librarians. There was nothing friendly about it. There were no home or office personal computers in those days, and anyone who used it, whether a computer professional or an engineer or scientist or librarian, had to learn to use a very complex system.

Fact Bytes



Vinton Gray Cerf, an American computer scientist is recognized as one of the **Father of the Internet**. Some of his important contributions were in:

1. Developing TCP/IP technology.
2. Development of the first commercial e-mail system.



How are so many computers connected?

The Internet uses special cables, telephone lines, satellites, microwaves and other devices to connect thousands of computers to each other.

The Internet has some powerful computers called **servers** which process the commands given by the thousands of connected computers called **nodes**.

Internet Terms

Home Page

A **home page** is the first page you see when you enter a website. The home page is linked to all the other web pages of the website.





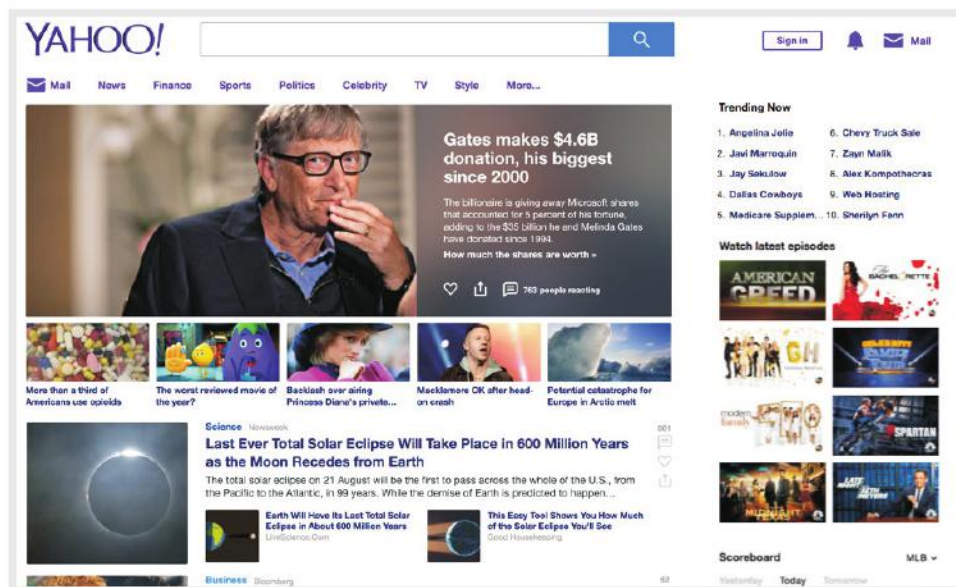
Search Engine

A search engine is a computer program that searches for information on the Internet. Google (<http://www.google.com>) is the world's most popular search engine. Some of the other popular search engines are listed below.

Yahoo Search(<http://in.search.yahoo.com>)

Wikipedia (<http://wikipedia.com>)

Bing (<http://www.bing.com>)



URL (Uniform Resource Locator)

Each website has an address which is called its **URL**. The URL refers to the address of the web server on which the website is stored.

Examples

<http://www.microsoft.com>

<http://www.howstuffworks.com>



Web Browser

To access the Internet you require a software called a web browser to be present on your computer. A web browser allows you to read the web pages on the World Wide Web.

Examples:



Fact Bytes



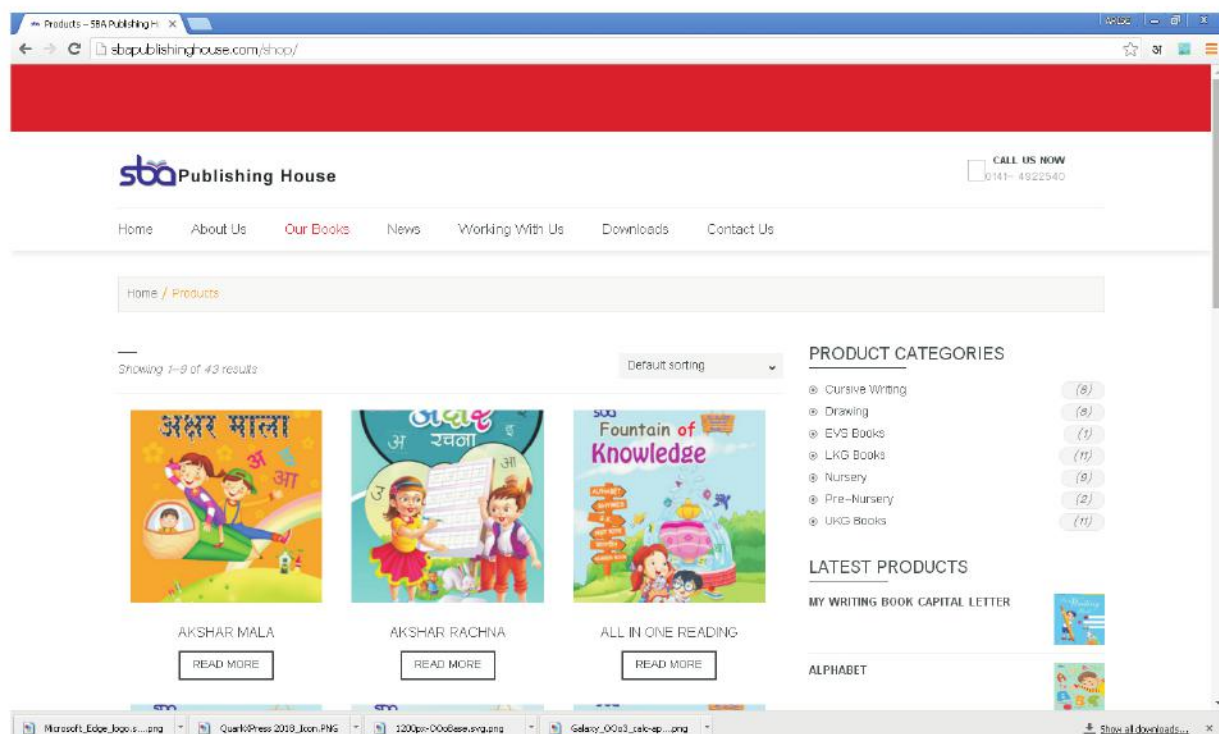
Internet Explorer is the default Windows browser in Window7. Mozilla Firefox and Google Chrome are two other very popular browsers.

Browsing is reading the web pages on the WWW using a web browser.

Web Page

A **web page** is an electronic document on the World Wide Web. A web page is written in a computer language called **Hyper Text Markup Language (HTML)**.

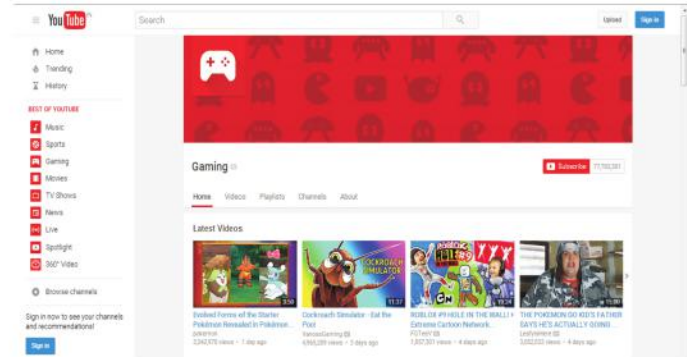
1. Web pages can contain text, graphics, animations, sound and even video.
2. They are connected to each other by linked words called **hyperlinks**. When we click on these words, we get linked to another web page or image or song or even a movie.
3. Web pages are designed by using HTML coding software like **Java script** and **Dreamweaver**.



Website

A **website** is a collection of web pages related to each other by a certain topic.

1. A website is created by an organization or a person to give information about a particular topic, person, product or organization.
2. Moving from one website to another on the net is known as **net surfing** or **browsing**.
3. Websites are stored on **web servers** located all over the world. Yahoo.com, Amazon.com, Indiatimes.com, etc. are different websites, each with different types of content.



World Wide Web (WWW)

World Wide Web is a collection of various websites. These sites collectively form a structure similar to a spider web over the Internet and are therefore called World Wide Web.



Web Server

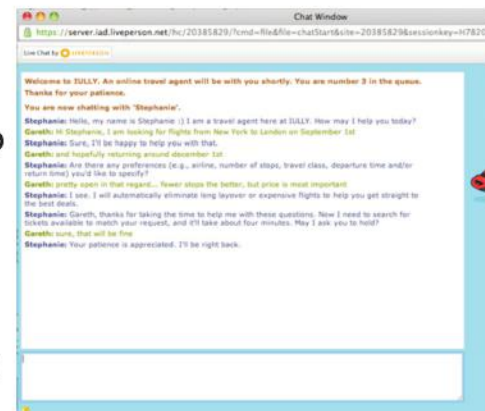
Web servers are big, powerful computers which store web pages and websites. When you enter the URL of a web page in your web browser or click on a hyperlink.

Going Online

When you are connected to the Internet and are on the World Wide Web, you are **online**. You can take part in many activities like surfing, watching movies, listening to music, emailing when you are online.

Uploading

Uploading a file is moving a page from your computer onto the World Wide Web. An uploaded file is stored on a web server. We need special software to upload files.



Downloading

Downloading is saving a page from the web to your computer's hard disk for future reference. We can download music, software, movies from the net.



Surfing

The process of moving from one website to another on the WWW is known as **surfing**. Like we flip through the pages of a magazine, we surf from one web page to another searching for information.

Requirements to Access the Internet

The Internet is not owned by a single person or organization. Anyone can become a part of this largest network of the world and can share the available information. To connect to the Internet, you need the following listed things:

1. A computer system with **Internet Browsing Software** such as Microsoft Internet Explorer or Google Chrome.



2. A **telephone line** that can be used for sending and receiving the data.



3. A **MODEM** (Modulator-Demodulator) to connect the telephone line with a computer.

It converts digital data of the sender's computer into analog signals that can be transmitted over the phone line and converts them back to digital form at the receiving end.



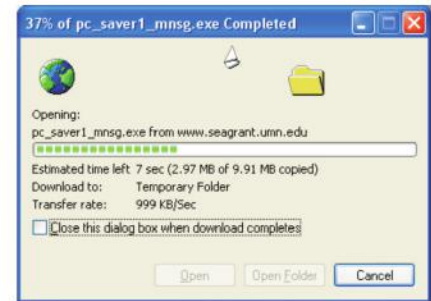
4. An **Internet Service Provider (ISP)** is a company that offers connectivity to the Internet. When we send or receive the information on the Internet, it first reaches the ISP. Some Indian ISPs are BSNL, VSNL, MTNL, Airtel, Tata and JIO.



Communicating over the Internet

People all over the world use the Internet to connect to each other. They do so by:

- Sending e-mails
- Chatting online
- Video conferencing



- ▶ Being part of news groups which are online discussion forums on various topics
- ▶ Social Networking in which people use online
- ▶ Social networks like **Facebook**, **Instagram**, **Yahoo Messenger** and **Whatsapp** to keep in touch with friends
- ▶ Putting up websites
- ▶ Blogging



Fact Bytes



Blogs are online journals maintained by people to give their opinions or share their ideas. Microblogging sites like twitter.com have millions of users all over the world.

E-mail

Over the years, the ways of communication have changed. E-mail has become the fastest and the most reliable way of communication.

E-mail means sending and receiving mails electronically using computers. It is similar to writing letters and is a quick way of keeping in touch with people.

Some websites provide e-mail service to send and receive e-mails. They are called e-mail service providers.

Some popular e-mail service providers are listed below:

- ▶ www.gmail.com
- ▶ www.yahoomail.com
- ▶ www.rediffmail.com
- ▶ www.hotmail.com

Advantages of E-mail

Fast : Messages can be sent anywhere around the world in an instant.

Cheap : Transmission usually costs nothing, or at the most, very little.

Simple : It is easy to use after initial set-up.

Efficient : An e-mail can be sent to a group of people in one step.

Versatile : Pictures, PowerPoints or other files can also be sent through e-mails.

Netiquette

Netiquette refers to a set of rules that should be followed while using the internet to respect the privacy of other people.

Following rules should be followed while using the internet:

- ▶ Write clearly so that other users are able to understand what you are trying to convey.



- Be very careful about what you write. Your posts are public and can be viewed by anyone.
- Avoid using all caps while writing. It is considered rude. It makes the others think that you are shouting at them.
- Use appropriate language while writing. Avoid spelling and grammatical errors.
- Make sure that the information provided by you is true in every aspect. However, do not share your personal information like your address and telephone number or banks details without your parent's permission.
- Always be yourself while doing any kind of online activity. Do not fake in anyway. Avoid using fake e-mail IDs.

Syntax of an E-mail Address

username@host name

Example: **abcd123@gmail.com**

where **abcd123** is the user name and **gmail.com** is the name of the host server.

The host name is the name of the e-mail server which is providing you the e-mail services.

Some of the popular e-mail sites are: www.gmail.com, www.yahoo.com, www.hotmail.com

Getting to Gmail

To send and receive e-mails, you need to have an e-mail account. Gmail is really a smart e-mail service that boasts not only of a huge and growing amount of e-mail storage but also offers an intelligent approach to messaging that always lets you stay abreast of your inbox and see the relevant e-mails.

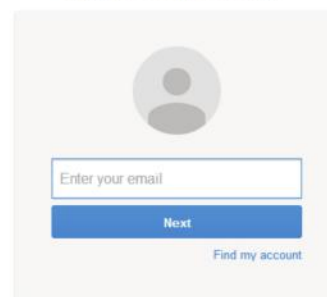
1. Type **www.google.com** in the address box. The Google homepage will open up. Click on **Gmail**. OR

Type **www.gmail.com** on URL, the **Gmail Sign in** page opens up.



One account. All of Google.

Sign in to continue to Gmail



[Create account](#)



Creating a New Gmail Account

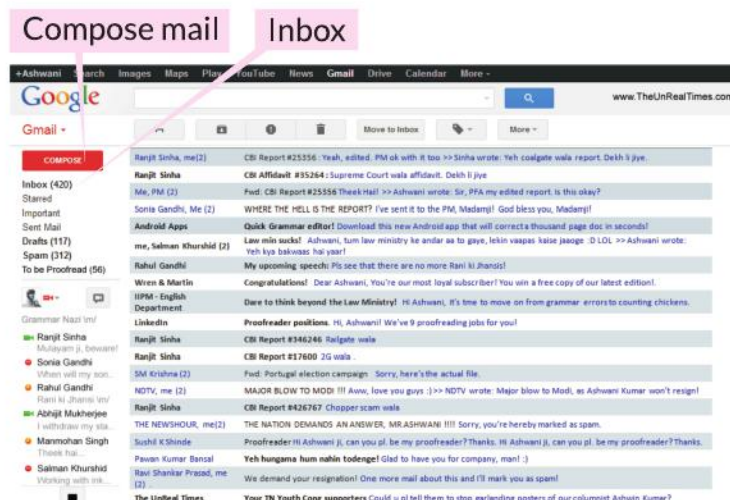
If you are new to Gmail and do not have an account, you need to create one.

Click on the **Create an Account** button. A registration form opens up. Fill in the details correctly by following all the instructions. Make sure you set a strong **password** for your account. Once you have submitted the form, you can operate your account.

Create your Google Account

To Use an Existing Account

Enter the **User name** and **Password** and click **Sign in**. Your G-mail account opens up as shown on the figure.



Inbox

The **Inbox** shows all the mail received by us. The senders name, subject, date, time are all displayed. To view an e-mail, we have to click on it.

Compose Mail

Click on **Compose** to write a new e-mail. The **compose mail** box opens up.

1. Type the e-mail address of the person we are mailing to.
2. Type the subject of the e-mail.
3. Type the contents of the e-mail.
4. You can make the text of your mail attractive by using the **formatting tools** given at bottom of the **compose mail** box.

5. You can send attachments such as pictures, documents, music and presentation etc. with your mail. But be careful that the size of the **attachment is within 25MB**.

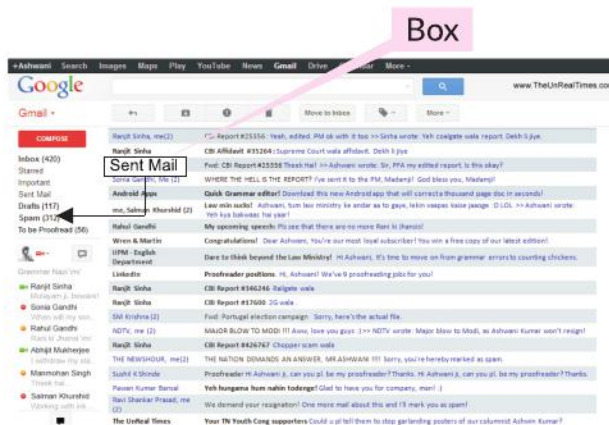
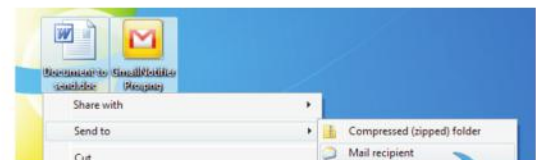
To send an Attachment

1. Click the **Attach file** button.
2. The Open window appears.
3. Select the file you want to attach by double clicking it.
4. The file is now attached to your e-mail.



Can you see the paper clip showing the attached file?

When your mail is ready to be sent click the **Send** button. The recipient will receive the mail immediately!



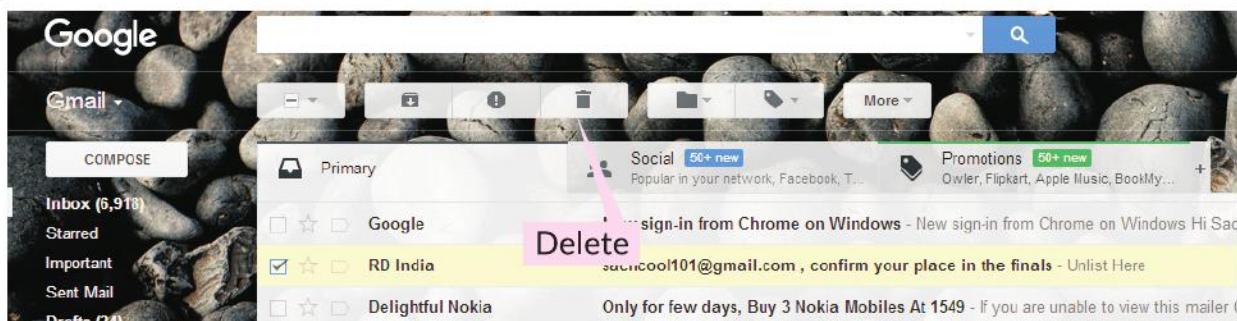
Attach File Button

Sent Mail

To check if your mail has been sent click on the **Sent Mail** button and check the sent mail in the Sent Mail window.

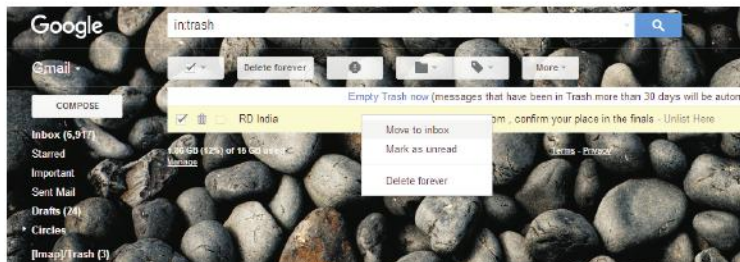
Delete Mail

To delete mail from your Inbox or Sent Mail, click the check box next to the mail and click the **Delete** button to delete the message. The mail has now gone into the **Trash** box.



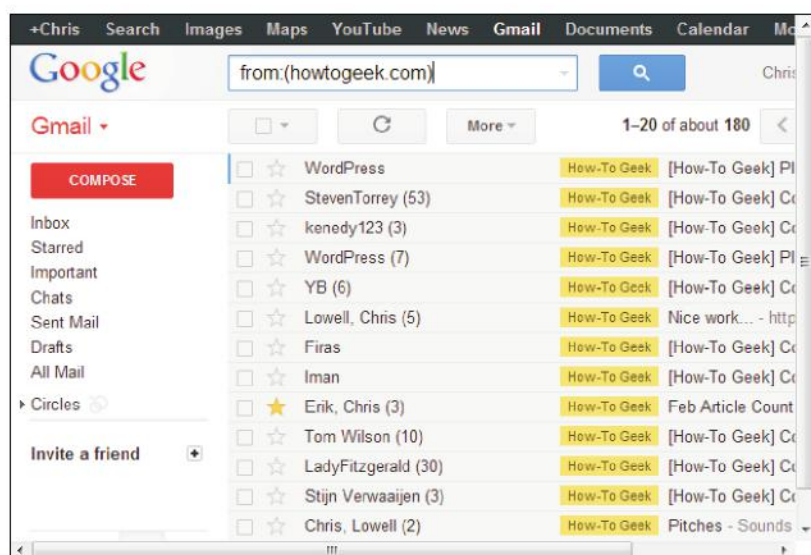
To Retrieve Mail From the Trash Box

1. Click on the **Trash** button. The Trash window opens up. Click the check box next to the mail which is to be retrieved.
2. Click on the **Move To** button. Select **Inbox**. The deleted mail will be sent back to the **Inbox**.



Search Mail

Use the **Search mail** text box to search for mail written by a friend or on a particular topic.



Sign Out

Click the **Sign Out** button of your account. You must always Sign Out of your account when you finish checking or sending mail.

Be Smart and Safe!



1. Keep safe by not revealing personal information while using the internet. It includes e-mail ID, house address, photos, phone number and password.
2. Meet online friends only in the presence of an adult family member.
3. Accept e-mails and messages only from people you know. Be aware of viruses and spam messages.
4. All the information on the internet may not be correct. Check the reliability of the information with other sources – books, websites or someone who knows.
5. Tell your teacher or an adult family member if you or someone you know is uncomfortable about an online communication.



Let's Summarise



- ▶ Internet is a global network of computers.
- ▶ www, webpage, website, Home page, etc. are some important terms related with Internet.
- ▶ Modem, ISPs and web browsers are required for getting connected to the Internet.
- ▶ People all over the world use the Internet to connect to each other via sending e-mail, video conferencing and chatting online.
- ▶ E-mail is the most widely used application of the Internet.



Practice Time

A. Tick (✓) the correct option.

1. _____ is the short form of electronic mail.
(a) E-mail ☐ (b) G-mail ☐ (c) Yahoo ☐
2. _____ is the collection of various websites.
(a) Website ☐ (b) Web Browsers ☐ (c) World Wide Web ☐
3. _____ is the address of a webpage.
(a) Hyperlink ☐ (b) URL ☐ (c) Home Page ☐

B. Write T for true and F for false statements.

1. A webpage is an electronic document on the world wide web. ☐
2. To access the Internet you require a software called web page. ☐
3. Web servers stores webpages and websites. ☐
4. The signout button is used to search the mails. ☐

C. Fill in the blanks.

1. _____ means sending and receiving mails electronically.
2. A _____ line is used for sending and receiving the data over the internet.
3. A _____ is used to connect the telephone line with a computer.
4. The first page of any site is called _____.
5. _____ a file is moving a page from your computer onto World Wide Web.

D. Answer the following questions.

1. What is the Internet and how does it evolve?



2. Explain the requirements to surf the Internet.

3. What is an e-mail? What are its advantages?

4. What is the role of keywords in Search Engine?

5. Write the steps of creating a gmail account?

Fun Time



► Identify the following web browser.



Lab Time



► Go online and write down the URLs of sites where you could do the following activities.

- Buy and sell things
- Buy books
- Study online
- Download music
- Get information about India
- Send emails

Brain Booster



► Which is faster— e-mail or postal mail?

Project Time



► Choose a topic of your interest from your course books (on any subject). Open a search engine (like Google). Now search for more information on the above topic and create a presentation on it.



Test Paper-I

Based on Chapters I-4



A. Tick (✓) the correct option.

- _____ is an example of operating system.
(a) UNIX ☐ (b) DBMS ☐ (c) MICR ☐
- _____ option aligns the text evenly between left and right margins.
(a) Left ☐ (b) Right ☐ (c) Justify ☐
- _____ is the most basic software.
(a) Operating System ☐ (b) Utility Software ☐ (c) Graphics ☐
- The group of eight bits is a _____.
(a) Nibble ☐ (b) Gigabyte ☐ (c) Byte ☐

B. Write T for true and F for false statement.

- The application software is designed to help the users to perform specific tasks. ☐
- Creating list are of two types. ☐
- The personalization of themes is not possible. ☐
- Flash drives are also known as Pen drives. ☐

C. Fill in the blanks.

- Aero is a type of pre installed _____.
- _____ is the default spacing between the lines of a paragraph in MS WORD.
- _____ is the full form of DVD.
- _____ software is used to design and create slideshows.

D. Answer the following questions.

- What is the use of primary memory?

- What is the difference between Bulleted and Numbered list?

- Write the steps of changing the date and time settings of your computer.

- Give details of any two utility software.



Test Paper-II

Based on Chapters 5-8



A. Tick (✓) the correct option.

- _____ appears at the top of MS Excel window.
(a) Title Bar ☐ (b) Ribbon ☐ (c) Formula Bar ☐
- What is the shape of Input/output box?
(a) Circle ☐ (b) Rectangle ☐ (c) Parallelogram ☐
- _____ is the collection of various websites.
(a) Website ☐ (b) Web Browsers ☐ (c) World Wide Web ☐
- Copy and _____ options are used to copy the text.
(a) Cut ☐ (b) Paste ☐ (c) Both (a) and (b) ☐

B. Write T for true and F for false statement.

- The Save As option is present in File Tab. ☐
- No modifications can be made to the predesigned themes. ☐
- Flowcharts are not important in computer programming. ☐
- Web servers stores webpages and websites. ☐

C. Fill in the blanks.

- _____ contains data in the form of rows and columns.
- _____ is the shape of processing box in flowchart.
- _____ means sending and receiving mails electronically.
- Clipboard group is present in _____.

D. Answer the following questions.

- Write the steps for saving a Excel workbook.

- Explain the text alignment options to align the text in the placeholder.

- What is an e-mail? What are its advantages?

- Write any two rules for writing an algorithm.

